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**Mobile application for behavioral risk
management associated with
cryptocurrency investing**

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Abstract

This thesis investigates the influence of behavioural economic biases on decision-making in cryptocurrency trading. Drawing on established theories from behavioural finance, specifically Prospect Theory, developed by Nobel laureate Daniel Kahneman and Amos Tversky, it identifies how various cognitive biases and heuristics, such as overconfidence, availability heuristics and loss aversion, impact decision-making under uncertainty. These theoretical foundations are directly applied to the context of cryptocurrency trading. To mitigate irrational decision-making, a mobile application was designed and developed to detect and mitigate behavioural biases in real time. The app was evaluated through user testing and thematic analysis, with findings indicating that the system reliably identifies key biases and provides meaningful feedback. However, challenges remain regarding beginner accessibility, coverage of additional biases, and internal security mechanisms. The study concludes by outlining areas for future development, including enhanced user onboarding, expanded feature sets, integration of additional bias detection, and the potential use of AI-driven algorithms for personalised interventions. The results demonstrate the practical value of behavioural economics in supporting safer and more rational participation in cryptocurrency trading.

Keywords

behavioural economics, prospect theory, cryptocurrency trading, cognitive biases, fomo, overtrading, overconfidence, loss aversion, bias detection, decision-making, risk management, financial decision-making, event-driven system

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List of Abbreviations

2FA Two-Factor Authentication.

API Application Programming Interface.

BFF Backend for Frontend.

CAQDAS Computer-Assisted Qualitative Data Analysis Software.

FOMO Fear Of Missing Out.

JWT JSON Web Token.

OTP One-Time Passwords.

P&L Profit and Loss.

REST Representational State Transfer.

SSE Server-Side Events.

Introduction

In the last decade, cryptocurrencies have moved from a niche interest for technology enthusiasts to a vital part of global finance. What began as a speculative hobby is now embraced by major economies: the United States has already established significant crypto reserves (The White House, 2025), while the Czech Republic is currently debating the establishment of its own multi-billion euro Bitcoin reserve (Reuters, 2025). At the same time, central figures in the U.S. are launching their own tokens (João da Silva, 2025).

This widespread enthusiasm and media buzz attract ordinary people to try crypto for themselves. Yet the features that can drive rapid gains, thinly regulated markets, and the potential for manipulation also put newcomers at risk of heavy losses. Lacking clear guidance, many individuals are vulnerable to cognitive biases, such as Fear Of Missing Out (FOMO), overconfidence or loss aversion, which can lead to irrational decisions and cause them to treat trading like gambling. As a result, they risk not only their savings but also their trust in the technology itself.

Understanding and addressing these risks requires more than traditional financial analysis. Behavioural economics, pioneered by Daniel Kahneman and Amos Tversky, studies how people make financial decisions in the face of uncertainty, systematically deviating from the "rational actor" model assumed in classical economics. Their impactful research, including the establishment of prospect theory, revealed that people consistently display cognitive biases like loss aversion and overconfidence (Kahneman, 2011; Kahneman & Tversky, 1979). Kahneman and Tversky's research was formally recognised when Kahneman received the Nobel Prize in Economic Sciences in 2002 for integrating psychological insights into economic science (Kate Whiting, 2024). Prospect theory remains a central framework in behavioural finance and is widely used to explain and predict the kinds of irrational trading behaviours seen in modern crypto markets (Chen et al., 2022). Thus, their contributions provide both the theoretical and empirical foundation for identifying and modelling the cognitive traps faced by crypto investors today.

Aims and motivation

To address these challenges, this thesis aims to design and implement a mobile application that would assist users in mitigating behavioural risks and the irrational decision-making that results from them. The application would detect behavioural biases in real time, drawing on the foundational work of Kahneman and Tversky as well as contemporary behavioural economics. Its goal would be to provide users with guidance to minimise the impact of irrational choices during cryptocurrency trading. The application should be accessible to beginners, offering an intuitive interface and clear explanations of financial concepts. Additionally, strong data protection and authentication measures are required to protect user privacy, in line with

the necessary standards for financial applications.

This project resonates with me for two reasons. First, cryptocurrencies are both timely and technically demanding: they challenge me to combine coding, financial analysis, and behavioural theory, which are areas I've long been passionate about, and so offer a chance to deepen my knowledge and skills. Second, I've seen more and more friends asking how to start investing in crypto and what pitfalls to avoid. By building this tool, I can help not only newcomers but also experienced traders navigate the crypto world more confidently.

Methods

To address the aims, this thesis utilises a multi-stage methodological approach that combines theoretical analysis, software development, and empirical user evaluation. The project began with a systematic literature review in behavioural economics and cognitive psychology, forming the conceptual basis for detecting and mitigating decision-making biases in cryptocurrency trading. Building on these insights, a software development process was undertaken to design and implement a cross-platform mobile application and supporting backend services. The effectiveness of the resulting system was then evaluated through qualitative user studies, including task-based usability testing and semi-structured interviews.

Work's structure

This thesis is organised into four main chapters, each addressing a key stage of the research and development process:

- **Chapter 1: Methodology** presents and justifies the methodological framework used throughout the thesis, including literature review, algorithm development, technical implementation, and user testing procedures.
- **Chapter 2: Behavioural Economic Theory and Its Application** reviews core concepts from behavioural economics, especially Prospect Theory, and adapts them to the context of cryptocurrency trading. This chapter also outlines the pseudocode algorithms used for bias and risk detection.
- **Chapter 3: Design and Implementation** describes the design decisions and implementation details of the developed application, including system architecture, backend services, risk detection modules, and the mobile application.
- **Chapter 4: User Testing and Evaluation** reports on the results of user evaluation, including participant feedback and thematic analysis of user experiences with the application.

Research Questions

To guide the design, development and evaluation of the proposed application, this thesis is structured around the following research questions. Each question directly supports the overall aim of creating a tool that helps users make more rational and secure decisions in the context of cryptocurrency trading:

- **RQ1:** How accurately do users perceive the app's behavioural bias detections relative to their own decision-making patterns?
- **RQ2:** How do users perceive and respond to the app's behavioural bias alerts in the context of decision-making?
- **RQ3:** How do users evaluate the app's security measures, and to what degree would they trust it for managing personal and financial data?
- **RQ4:** How comfortable and intuitive do users find the app's design and functionality during typical use?
- **RQ5:** How accessible and user-friendly is the app for individuals with little to no previous experience in finance or behavioural tools?

Limitations

During initial research, I identified a key constraint that will shape the future development of this proposed application. This limitation is the reliance on free, public-only APIs for market data. While this would keep the application free of costs, it also brings strict rate limits, lowered data granularity, and occasional gaps compared to enterprise-grade APIs. In a full implementation, paid data sources could improve accuracy and simplify integration, but would also introduce API costs.

Furthermore, the evaluation study's findings are limited by a small, relatively homogenous participant group, all of whom completed testing with demo accounts and no real financial risk. These factors may have influenced user behaviour and risk perception, and participation or interpretation biases cannot be fully excluded.

1. Methodology

This chapter outlines the methodological approach adopted to achieve the aims of this thesis: the development, implementation, and evaluation of a mobile application for mitigating behavioural risks in cryptocurrency trading. The methodology was designed to integrate both theoretical and practical perspectives, drawing from behavioural economics, user-centred design, and software engineering practice. The process began with a systematic review of relevant academic literature to inform the conceptual foundations of the application. This was followed by an iterative development phase, utilising a multi-language technology stack to build the core system. Finally, the application was evaluated through qualitative user studies, employing structured usability testing and thematic analysis of participant feedback.

1.1 Algorithms Development Process

The development process started with a systematic literature review conducted using scholarly databases such as **Google Scholar**, **Web of Science** and **ScienceDirect**. Selection criteria included relevance to behavioural economics, Prospect Theory, and cognitive biases in financial decision-making contexts. This theoretical foundation directly informed the iterative development and refinement of the algorithm, involving simultaneous literature review, theoretical synthesis, and practical coding.

ChatGPT 4.1 (OpenAI, 2024) was employed for stylistic corrections and as a research assistant for introductory overviews of behavioural theories using the Deep Research feature.

1.2 Technical Implementation and Choice of Technology

Python (v3.11) was selected as the primary programming language for its rich support for data analysis libraries such as **NumPy** (Harris et al., 2020) and **Pandas** (McKinney, 2010), making it well-suited for the required analytical tasks. Type validation was provided by **Pydantic** (v2.6.1) library. Market data was provided via official Bybt API connector **pybit** (v5.6.0). Messaging and event streaming were implemented with **confluent-kafka** (v2.8.2). **Pytrends** (v4.9.0) library was used to retrieve historical search interest for computing hype scores.

Java 21 with **Spring Boot** (v3.4.4) was used for the Backend for Frontend (BFF) server to coordinate data flow between services and the frontend, due to its suitability for such functionality and the author's prior experience with the framework. The backend leverages **Spring Data JPA** for database access, **Flyway** (v11.7.2) for schema migrations, and integrates security using **Spring Security** and JWT support via the **spring-security-oauth2-jose** (v3.4.4) module. Sensitive data, such as API keys, is encrypted via the **javafx.crypto** package.

Server leverages real-time communication through `Spring WebSocket`. `MapStruct` (v1.6.3) and `Lombok` (v1.18.32) are used for data mapping and code reduction. External trading APIs are accessed through the official `Bybit` (v1.2.7) Java connector.

The core trading engine, developed by the supervisor, was implemented in `Rust`.

The mobile application was built with `React Native` (v0.76.7) and `TypeScript` (v5.3.3) for cross-platform support. State management was implemented via `Redux Toolkit` (v2.6.1), and navigation was handled by `React Navigation` (v7.0.14). For security, biometric authentication and secure local storage were provided by `expo-local-authentication` (v15.0.2) and `expo-secure-store` (v14.0.1), respectively. HTTP requests were performed using `axios` (v1.6.7), while real-time data updates were handled using a native Server-Sent Events (SSE) connection via `react-native-sse` (v1.2.1). The project was bootstrapped and managed with `Expo` (v52.0.37).

`PostgreSQL` (v16.8) served as the primary relational database, with `Redis` (v8.0.2) providing an in-memory caching layer. `Kafka` (v4.0.0, Bitnami distribution) was used as the message broker between services.

Backend services were containerised using `Docker`(v27.5.1) running under `WSL2` (v2.5.9) with `Debian 12`, ensuring consistent and reproducible development and deployment environments.

Development was conducted using `IntelliJ IDEA`, `WebStorm`, `PyCharm` and `Visual Studio Code`.

GitHub Copilot (GitHub, 2024) was used for code autocomplete throughout the software development process. Claude Sonnet (Anthropic, 2024) was used as a code review assistant to help identify bugs and improve code quality. All AI-generated outputs were reviewed, validated, and edited by the author.

The source code was managed with `Git` and hosted on `GitHub`. All links to source code can be found in Appendix [A](#)

Additionally, video recording of app's features, or simply demo, can be found in Appendix [B](#)

1.3 User Testing and Evaluation

To evaluate how users interact with the application and perceive its core features: behavioural bias detection, prevention of irrational decision-making, security, and usability, a qualitative approach was selected. Data were gathered through post-use interviews, which were transcribed, coded, and analysed thematically using the six-phase model proposed by Braun and Clarke (2006).

The evaluation was structured around the research questions, defined in the Introduction (see p. [9](#)).

1.3.1 Evaluation approach

The evaluation was conducted in three phases: a task-based testing phase (following best practice in usability research (Nielsen, 1994; Rubin et al., 2011)), an optional free-roam period, and a post-use interview with open-ended questions. In the initial phase of the evaluation, each participant was asked to complete a set of structured tasks in the application. This approach is widely supported and helps identify concrete usability issues that may not emerge through abstract questioning or free exploration of the app (Nielsen, 1994). Rubin and Chisnell Rubin et al. (2011) argue that the task-based approach is especially effective for evaluating how well specific features work. This is important when assessing a new application, hence the choice of this approach. The chosen tasks were designed to cover all major functional areas of the application, including but not limited to account creation, security features, limit settings, job configuration and risk functionality, thus following best practices in task-based usability research (Lazar et al., 2017). Each participant was given a structured task instruction sheet at the start of the study. Participants were encouraged to complete all the tasks and note points of confusion or failure, and these notes were later discussed during interviews. After completing all the tasks, each participant was given one to three days of free-roam use, where participants could use the app on their own will, thus it was optional. All participants preferred to conduct testing with demo exchange accounts. The full instruction sheet is included in the Appendix C.

1.3.2 Sampling

All participants were recruited via convenience sampling from personal, professional and academic networks. This approach aimed to capture a range of financial and cryptocurrency experience levels in order to evaluate the application’s impact across diverse user backgrounds.

1.4 Participant Overview

Table 1.1 summarises key background and demographic information about participants:

Participant	Age	Gender	Occupation	Finance/Crypto Experience	Behavioural Finance Familiarity
P1	21	Male	Developer in bank	Bought and traded stocks, spot	Heard about it
P2	25	Male	Student (IT)	Bought and traded spot (occasionally)	Not at all
P3	22	Female	Student (Medicine)	Bought and traded spot via P2P	Heard about it

Table 1.1*Participant Demographics and Background*

1.4.1 Interview Process

The interviews followed a semi-structured format, which is considered an effective method to gather in-depth and reflective feedback while maintaining consistency across all the participants (Brinkmann & Kvale, 2015). Each interview was structured around a set of open-ended questions, designed to explore user perceptions and experiences across all major aspects of the application. The interview questions were divided into six thematic categories, corresponding to the key functional and conceptual areas of the app:

- **Behavioural Bias Detection & Impact** – assessing whether users noticed bias detection, found it relevant, and whether it helped to prevent irrational decision-making;
- **Security, Privacy & Trust** – evaluating user trust in the app and its data handling, and their perception of implemented security features;
- **Design & User Experience (UX/UI)** – understanding how users experienced the app visually and functionally;
- **Usability & Learning Curve** – examining whether the app was intuitive and accessible, especially for less experienced users;
- **Real-World Applicability & Future Use** – whether users anticipated integrating the app into real decision-making scenarios, considering future use of the app;
- **Overall Feedback & Suggestions** – general reflections, pain points, and areas for improvement.

Before the main interview began, participants were asked to confirm that they had completed the task-based phase and submitted the background information form. With consent, each interview was audio-recorded under the condition that all respondents would be anonymised, in accordance with ethical guidelines for human research (Oates et al., 2021). To establish comfort and ease participants into the discussion, a brief warm-up phase preceded the main questions, as they are commonly used to create a conversational tone and connection (Longhurst, 2003). This included general questions such as whether the participant had used any interesting apps recently (related to finance or otherwise), and how the participant would

describe their general approach to risk and decision-making. The complete set of questions is provided in Appendix [D](#).

Each interview lasted for approximately 30 minutes.

1.4.2 Data Analysis

After the interviews were completed, all audio recordings were transcribed using the `OpenVINO` (Version 2025.1.0) large model, which is an open-source speech-to-text transformation model. While this automated method significantly accelerated the transcription process, all transcripts were afterwards reviewed and manually corrected to ensure accuracy. Particular care was taken during corrections to preserve participants' original wording and emotional expression. In qualitative analysis, retaining the emotions of the participant is critical for maintaining interpretive reality. As Brinkmann and Kvale ([2015](#)) note, emotional tone, hesitations, and informal phrasing often express meaning beyond the literal content, and overly sanitised transcripts may suppress key aspects of it. Therefore, corrections were minimal, focused exclusively on improving readability without compromising context and emotional tone.

Full transcripts for participants 1, 2 and 3 can be found in Appendix [E](#), [F](#) and [G](#), respectively.

Finalised transcripts were imported into `Quirkos` (v3.0), a CAQDAS tool specifically designed to support the organisation, coding, and visualisation of thematic structures. `Quirkos` allows identifying co-occurrence patterns, comparing codes across participants, and supports visual outputs, enhancing transparency and analytical depth (Ting et al., [2024](#)).

The coding process followed a hybrid approach: combining inferential coding, using a predefined set of codes based on the research questions, with inductive coding, where new codes emerged organically from the data (Fereday & Muir-Cochrane, [2006](#)). This dual strategy allowed the evaluation to remain grounded in the study's objectives while also staying open to unexpected insights expressed by participants.

All coding was conducted in accordance with Braun and Clarke ([2006](#)) six-phase model of reflexive thematic analysis, which includes:

1. Familiarising with the data
2. Generating initial codes
3. Searching for themes
4. Reviewing themes
5. Defining and naming themes
6. Producing the report

2. Behavioural Economic Theory and its Application

Traditional economics generally assumes investors are rational decision-makers who consistently act to maximise their expected utility (Mankiw, 2016). However, real-world observations repeatedly challenge this assumption, revealing consistent deviations from rationality (R. H. Thaler, 2015). These deviations occur because human decision-making is heavily influenced by psychological biases, emotions, and cognitive shortcuts known as heuristics (Kahneman, 2011).

Behavioural economics developed with the aim of addressing these systematic deviations, leveraging psychological insights to form a more precise and realistic comprehension of economic behaviour (Ariely, 2009). By recognizing that human decisions frequently diverge from strictly rational computations, behavioural economics provides a powerful theory for understanding and predicting decisions in uncertain, volatile environments such as cryptocurrency markets (Almansour, 2020).

Among the many frameworks in behavioural economics, Kahneman and Tversky's Prospect Theory stands out due to its robust empirical validation and its direct applicability to financial decision-making contexts (Kahneman & Tversky, 1979). Prospect Theory systematically explains how people perceive risks and rewards differently based on reference points, how they overweight small probabilities, and why losses psychologically affect individuals more deeply than equivalent gains (Kahneman, 2011).

This chapter uses Kahneman's and Tversky's work as a foundation and shows how their theories and findings can be used to better understand trading decisions in the crypto market. The next sections explain how ideas from Prospect Theory and related behavioural studies can be turned into practical algorithms that help spot irrational trading behaviour.

2.1 Two Systems of Thinking in Kahneman's Study

Kahneman (2011) suggests that people think through an "automatic system" and an "effortful system", which are referred to as System 1 and System 2, respectively. System 1 is the fastest, automatic way of thinking. It operates with little to no effort and is used by the brain to automate decision-making in ordinary life based on past experience. This system was developed evolutionarily and is quite advantageous for humankind - it allows humans to react to threats and make decisions in a split second, which is crucial for survival. Kahneman proposed that System 1 relies on *heuristics* - the brain's mental shortcuts to process vast information and make difficult judgments (Kahneman, 2011).

This makes System 1 highly efficient, allowing it to make decisions without the brain's awareness. However, these shortcuts often cause predictable biases, which will be reviewed later. In contrast, System 2 is our brain's slow, analytical way to process information. System 2 requires effort and attention, so it is primarily engaged when people need to solve complex problems or make serious choices. Still, it is used much less often because it is mentally taxing, and the brain tends to save energy. It is important to notice how exactly System 1 and System 2 work together: System 1 runs automatically, and System 2 runs in a low-effort mode; when everything runs smoothly, System 1 continuously suggests answers to System 2, which accepts them with little or no modification. However, when System 1 fails to provide an answer, System 2 mobilises and provides a deep analysis (Kahneman, 2011).

To illustrate how these systems work, the following questions can be used:

- What is the result of 2×2 ?
- How many occurrences of the letter a on this page?

In these examples, the first question is a simple mathematical calculation, which is likely handled by System 1 automatically. For the second question, however, System 1 fails and engages System 2, which requires more focus and attention.

It is necessary to notice that System 1 and System 2 are adaptive. Initially, when the brain encounters a new, unique problem with no pre-existing associations in System 1, it relies on System 2 to handle the task. However, as the problem is repeatedly faced, the brain forms associations and patterns. This process allows System 1 to take over and handle the task more efficiently, using the learned associations. Over time, System 1 becomes capable of dealing with tasks that previously required System 2 (Kahneman, 2011). For instance, experienced traders, who are highly familiar with the trading process, are more likely to rely on System 1 when making decisions, which uses heuristics and results in a higher probability of biases. Heuristic-driven decisions are fast and automatic but can lead to judgment errors when faced with uncertainty or complex conditions. This is a crucial concept when designing a risk mitigation module. To better understand the biases that arise from System 1, it is needed to consider how people make decisions under uncertainty. Prospect theory, developed by Kahneman and Tversky, provides a framework for understanding how people evaluate gains and losses. It explains why decisions under such conditions might be irrational.

2.2 Prospect Theory: Core Concepts

Prospect Theory, introduced in the paper "Prospect Theory: An Analysis of Decision under Risk" (Kahneman & Tversky, 1979), is a comprehensive framework describing how people make choices when evaluating potential financial outcomes. It explains how individuals perceive risk and rewards differently, often leading to irrational decisions. There are three key principles, which are described in the following subsections:

1. Evaluation relative to a Reference Point
2. Overweighting small probabilities
3. Loss aversion

2.2.1 Evaluation Relative to a Reference Point

One of the cornerstones of Prospect Theory that explains how people perceive risk is the reference point. The reference point describes the concept that the carriers of value are changes in the value, not the final state of the value itself (Kahneman & Tversky, 1979). According to this principle, people's perception is based on differences in value relative to a reference point, and this applies to both sensory attributes (e.g. room temperature) and nonsensory attributes, such as wealth or well-being. A reference point can be a previous state, expectation or any other benchmark.

The concept of reference points seamlessly applies to crypto trading, where traders do not evaluate their assets in isolation. Instead, their reference point is often the purchase price of a coin, a previous peak of the price or a price expectation of an individual trader, and the reference point is usually represented by candlestick patterns or simple moving averages (Bitpanda, n.d.).

For example, a trader who bought Bitcoin at an initial price of \$50,000 will perceive the new price of \$60,000 as a gain relative to the initial price. If the market price drops to \$40,000, it will be perceived as a loss if the reference point was set at the initial price. However, if a trader bought the coin at \$50,000, but expected the price to rise to \$70,000, a price of \$60,000 will be perceived as a loss, since it is lower than his reference point. This is important for understanding further concepts.

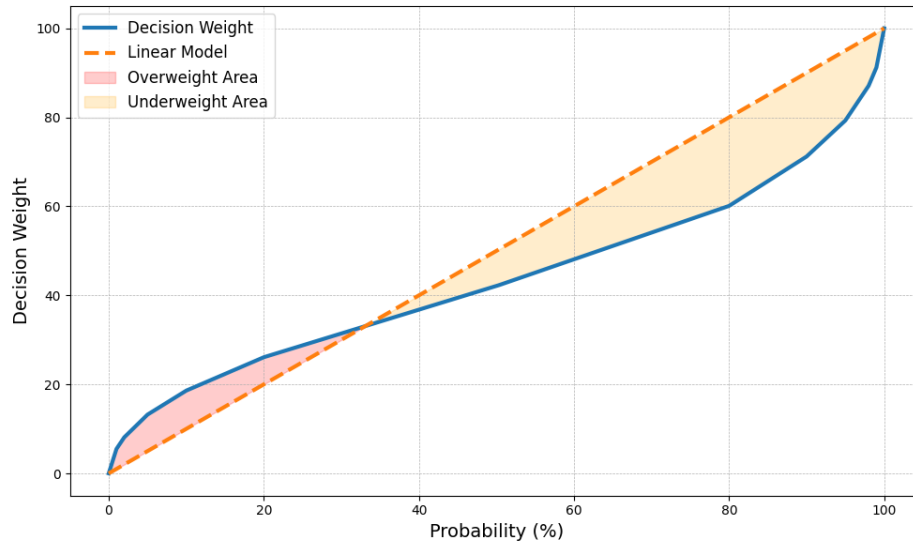
2.2.2 Overweighting Small Possibilities

Overweighting small possibilities is another principle of Prospect Theory, and it refers to the tendency of individuals to overestimate the probability of rare events while underestimating the likelihood of more common events. This phenomenon challenges the traditional economic assumption that people evaluate probabilities solely based on numerical value. Instead, Kahneman suggests that individuals exhibit irrational decision-making and that people assign decision weights to probabilities. It is important to notice that in certain extreme probabilities, like 0% or 100%, weights correspond to the probability itself, assuming individuals treat them rationally (Kahneman, 2011).

Probability (%)	0	1	2	5	10	20	50	80	90	95	98	99	100
Decision weight	0	5.5	8.1	13.2	18.6	26.1	42.1	60.1	71.2	79.3	87.1	91.2	100

Table 2.1

Decision weights for probabilities (Kahneman, 2011)

**Figure 2.1**

Graph showing the relationship between probability and decision weight, based on the data from (Kahneman, 2011)

This can result in irrational decisions, such as gambling on low-probability events or overlooking potential risks or disasters. In crypto trading, this concept might lead to traders speculating on low-cap, low-liquidity tokens, seeking large and quick profit. Traders often believe these tokens might experience a sudden price increase, similar to the rise of Bitcoin or altcoins. Traders overestimate this possibility, even if such an outcome is extremely low.

A classic example of such behaviour often revolves around "meme" coins, which are easy to create and primarily driven by speculative interest (Turk, 2025). Dogecoin, one of the most notable meme coins in the market today, is a great example. Originally created as a joke, Dogecoin experienced a significant increase in popularity and market capitalisation, peaking at 88 billion USD (Nani, 2022). But Dogecoin is rather an exception than a standard. Binance, one of the major crypto exchanges, conducted research and found out that 97% of "meme" coins fail due to Pump-and-Dump schemes, lack of value or speculation (Joshua Wong, 2024). Investing in such coins is risky because such investments are often characterised by low liquidity, making it difficult to sell or exit positions.

Given the high-risk nature of such and similar coins, it is important for traders to implement effective risk management strategies. The theoretical risk mitigation module could help traders avoid risks by notifying them if it detects any asset that might be subject to market manipulation, evaluating the asset's market liquidity.

Source code 2.1

Pseudocode example overweighting

```
\\ Pseudocode
if asset.liquidity < THRESHOLD and asset.volume < THRESHOLD
```

```
return "Warning: you are opening positions with low-liquidity coins,
        overweighting the chance of their future success"
```

Where:

- **asset.liquidity < THRESHOLD**: Comparing asset liquidity with the threshold, evaluating if it's high enough.
- **asset.volume < THRESHOLD**: Comparing amount invested into the asset with defined threshold.

2.2.3 Loss Aversion

Loss aversion, the key idea of Prospect Theory, demonstrates the psychological asymmetry between gains and losses. In one of the experiments that Kahneman and Tversky conducted, participants were asked to choose between two prospects that involved either a certain gain or probabilistic gambling for a higher gain. People typically preferred the guaranteed gain, even when the expected value of the gamble was higher. This pattern of behaviour demonstrates the psychological asymmetry between gains and losses. As Kahneman stated himself - 'losses loom larger than gains' (Kahneman & Tversky, 1979, p. 279). Loss aversion suggests that individuals generally go to greater lengths to avoid losses than to achieve gains of the same size.

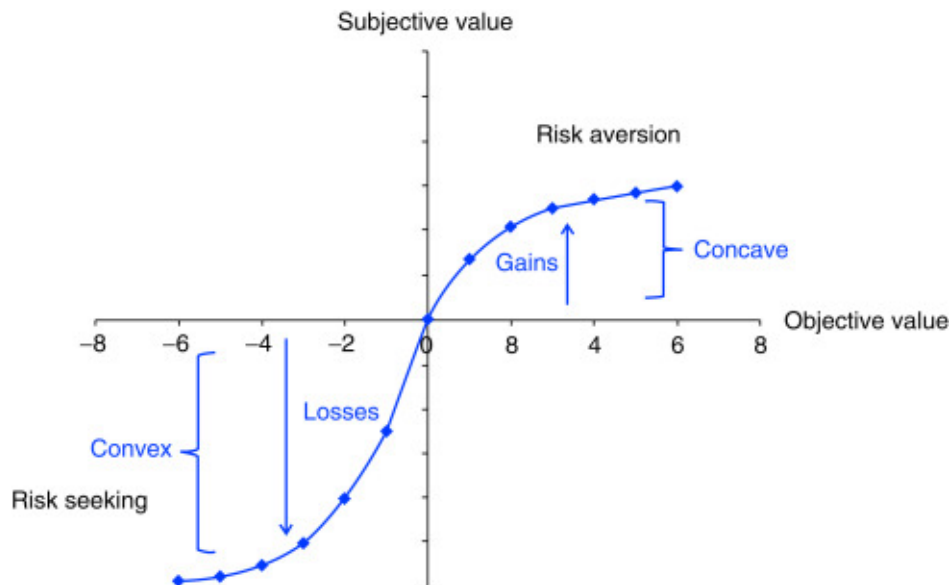


Figure 2.2

An Illustration of a Value Function (Abramson, 2016).

This defines two types of behaviour: individuals may become risk-averse when facing potential gains and risk-seeking when faced with potential losses. Kahneman estimated the ratio at which risk-aversion behaviour turns into risk-seeking as approximately 1.5 - 2.5 of the

reference point (Kahneman, 2011).

In trading, and especially crypto trading, where market conditions are volatile, loss aversion can lead to significant consequences. Traders holding losing positions might hesitate to close them because experiencing losses is psychologically more impactful. This discomfort often results in risk-seeking behaviour, where traders avoid realising the loss. In such a case, traders might engage in sunk cost fallacy or double down, biases where a trader continues investing in a losing position, even when accepting the loss and the liquidation of positions is the only rational decision.

Alternatively, loss aversion also impacts winning positions. Traders holding winning positions might close them as soon as the price rises above their reference point, often perceiving this change as a gain. This is risk-aversion behaviour, which arises from the desire to secure profit. While this action technically locks in a profit, it often leads to underperformance over time. By closing positions prematurely, traders miss out on potential further gains when market conditions suggest a further rise in the price. Both of these behaviours are explained by loss aversion and the disposition effect, which suggests that individuals have a tendency to sell winning positions over losing ones. This contributes to underperformance in long-term investing, as traders fail to capitalise on the full potential of their profitable positions while holding unprofitable ones (Kahneman, 2011).

A potential risk-mitigating module could help detect when traders exhibit risk-seeking and risk-averse behaviour based on their actions. Such a module can warn traders when they might show risk-seeking behaviour by holding onto losing positions for too long:

Source code 2.2

Pseudocode example risk-seeking

```
\\ Pseudocode
if position.unrealized_pnl <= -10% and position.holding_time > THRESHOLD
    return "Warning: you hold losing positions for too long, showing
           risk-seeking behaviour"
```

Where:

- **position.unrealized_pnl <= -10%:** Position's unrealised profit is 10% or more negative.
- **position.holding_time > THRESHOLD:** Checks how long the position has been held. If the holding time exceeds a threshold, it assumes that the trader has been in a losing position for an extended period.

or the module could detect when a trader secures profits too early, showing risk-aversion behaviour:

Source code 2.3*Pseudocode example risk-averse*

```
\\ Pseudocode
if position.realized_pnl <= 5% and trader.cut_positions > 3
    return "Warning: you might be cutting profits too early, showing
           risk-aversion behaviour"
```

Where:

- **position.realized_pnl <= 5%:** Position's realised profit is less than 5% .
- **trader.cut_positions > 3:** Checks if the trader has more than three cut positions, assuming this is a tendency.

2.3 Availability Heuristic

The availability heuristic is a cognitive bias that affects people's decisions based on how effortlessly examples come to mind. This mental shortcut often leads to overestimating the likelihood or importance of an event if associations tied to the event are recent, vivid, or emotionally charged (Kahneman, 2011). In Tversky and Kahneman's seminal paper on the availability heuristic, they conducted several examinations to demonstrate how this heuristic influences decision-making. One of the most illustrative experiments involved 2 groups of participants, consisting of 87 and 114 individuals, respectively. Both groups were then shown an expanded factorial of eight, which they had to solve within **five seconds**.

The first group was shown the following descending sequence:

$$8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$$

The second group was shown the ascending sequence:

$$1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7 \times 8$$

Although the correct answer is 40,320 for both expressions, the groups estimated the product differently. The first group's estimated median of the product was 2,250, while for the second group it was 512. Kahneman and Tversky explained this difference in the estimations by how people's brain processed the sequence. Both groups, due to limited time, extrapolated from a partial computation of **easily available information**, which in this case were first number in the sequences: the first group, which was shown descending sequence, started their estimation with larger numbers ($8 \times 7 \times 6$), which resulted in a higher estimate. Contrarily, the second group, presented with the ascending sequence, started with smaller numbers ($1 \times 2 \times 3$), leading to a lower estimate. This shows how the availability heuristic can affect people's decision-making based on the availability of information (Tversky & Kahneman, 1973).

The availability heuristic plays a significant role in crypto trading, where traders' decisions might be influenced by recent and high-profile news events. In the volatile nature of the

crypto market, traders tend to rely on easily accessible information, such as breaking news or sensational headlines, to guide their decisions. For example, when China announced its ban on cryptocurrency trading on May 18, 2021, the news rapidly spread across global media and social platforms. Given China's position as one of the world's largest economies, many traders might have assumed that other countries would follow this trend, influenced by the recent example that was in the traders' minds. On the day of the announcement and the following day, the market experienced a price correction across most of the coins and a spike in volatility. This sudden consequence was likely driven by panic selling, withdrawals and the emotional reaction of the traders. However, the volatility spike and the price correction were short-term trends, as over a 40-day timeframe, volatility returned to almost pre-event value (Manwaring, 2021).

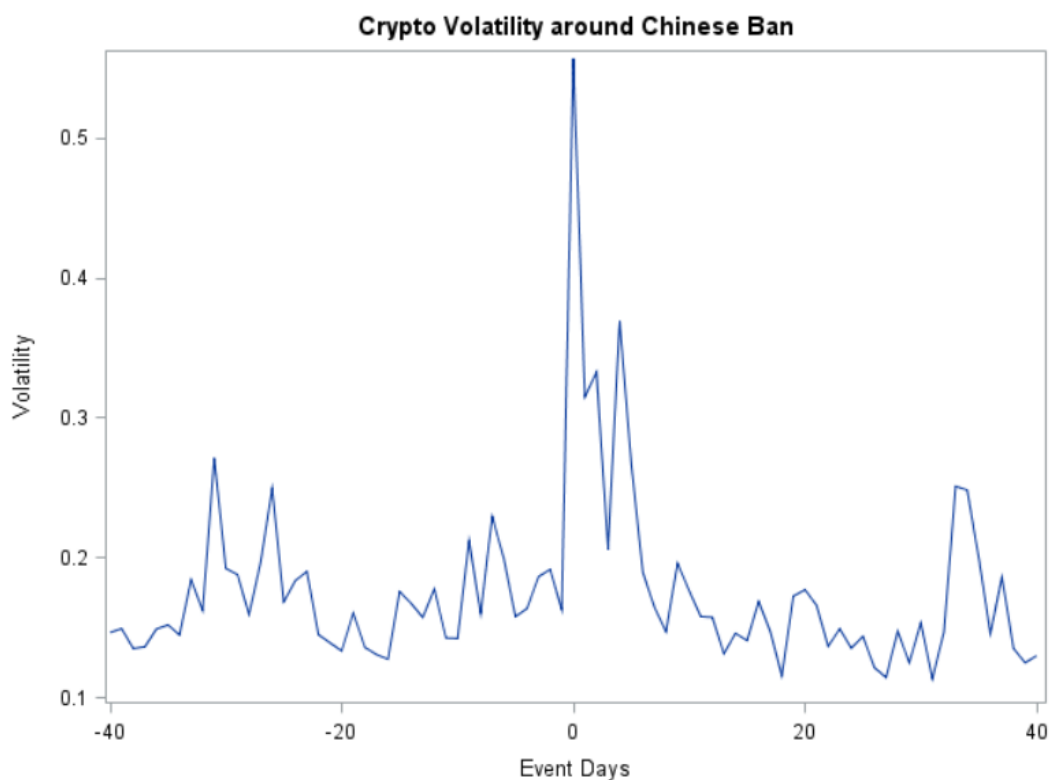


Figure 2.3

Volatility around China's ban on cryptocurrencies (Manwaring, 2021).

The short-term nature of these market impacts further indicates that they were driven not by a shift in the market's fundamentals but by the influence of recent news, which aligns with the principle of the availability heuristic. This effect can also significantly drive the Fear Of Missing Out (FOMO) effect. In these situations, the availability heuristic leads traders to overestimate the importance of an event. At the same time, FOMO pushes them to act impatiently, driven by the fear of missing a potential opportunity.

A potential risk mitigating module can help traders avoid the consequences of the availability heuristic. While it is difficult to directly estimate whether a trader's actions are influenced

by recent news or events, certain market and news data patterns can help indicate short-term market fluctuations and warn the trader. For example, the module could monitor coin volatility, warning a trader if a coin is unstable at the moment:

Source code 2.4*Pseudocode example availability heuristic*

```
\\ Pseudocode
if recent_volatility_change >= THRESHOLD
    return "Warning: this asset is experiencing fluctuations."
```

Where:

- **recent_volatility_change >= THRESHOLD**: Checks if volatility increased more than the threshold/baseline, assuming it is a common pattern for the market fluctuations.

Additionally, the module could monitor recent news count for coin and warn if a sudden increase is detected, assuming it might involve availability heuristic's behaviour:

Source code 2.5*Pseudocode example availability heuristic 2*

```
\\ Pseudocode
if news_count >= news_count_baseline
    return "Warning: this asset is experiencing increased attention in the
           news, it might affect its price"
```

Where:

- **news_count >= news_count_baseline**: Comparing today's news count with the average news amount of the same coin.

2.4 Overconfidence

Overconfidence bias refers to the tendency of people to overestimate their own knowledge, abilities or skills to control the outcomes of their actions. This bias results in people believing that they are more skilled or better informed than they actually are, often leading to riskier decisions (Kahneman, 2011). People overestimate their own commitment to past positive outcomes, recalling information about their success more easily than failures. This leads to individuals having unrealistically positive self-evaluations (Odean, 1998). For example, 81% of new business owners in the US thought their business had a 70% or better chance of success, whereas only 39% of the same owners thought that others had the same chance

(Cooper et al., 1988).

An illustrative example of overconfidence and its negative impact in finance was found in research conducted by Barber and Odean (2000), who analysed the trading behaviour of 66,465 households in the United States. Their study revealed that the most active traders (by monthly turnover), specifically the top 20% of the traders, significantly underperformed compared to the less active traders, reaching up to 10,3% lag of annual net returns (Barber & Odean, 2000).

Barber and Odean assumed that this lag was partly a result of excessive trading, which led to higher transaction costs and lower returns. They argue that this overtrading was driven by overconfidence, where traders overestimated their ability to predict the market. This highlights the harmful effects of overconfidence on trading behaviour.

The nature of overconfidence is similarly evident in crypto trading, but is often even more forceful due to the high volatility of the market. Similar to the findings in Barber and Odean's study, overconfidence in the crypto market often results in overtrading, which is characterised by a large number of trades, high trading volumes, frequent portfolio adjustments, or not utilising any safety mechanisms, like stop-loss. This excessive trading might lead to stagnated returns, as the trader's profit is offset by transaction costs or even to significant losses.

A potential risk mitigation module could detect the overconfidence bias primarily by monitoring its main behaviour pattern, which is overtrading. By evaluating parameters like daily trade amount and daily volume of trades, the module can warn traders if they show signs of overtrading:

Source code 2.6

Pseudocode example overconfidence

```
\\ Pseudocode
if daily_trade_amount >= THRESHOLD or daily_trade_volume >= THRESHOLD
    return "Warning: you are showing signs of overtrading"
```

Where:

- **daily_trade_amount >= THRESHOLD:** Comparing the trader's trade count per day with a threshold, which secures a pattern of opening many small trades.
- **daily_trade_volume >= THRESHOLD:** Comparing the trader's trade volume per day with a threshold, which secures a pattern of opening a few high amount trades.

Additionally, for stricter control, the system can evaluate a time gap between trades and warn if it is small, assuming a trader is under the influence of overtrading:

Source code 2.7

Pseudocode example overconfidence 2

```
\\ Pseudocode
LIST trades;
if (trades.getLast().timestamp - trades.getLast(-1).timestamp) < THRESHOLD
    return "Warning: time gap between trades was not fulfilled, you are
           showing signs of overtrading"
```

Where:

- **trades.getLast().timestamp - trades.getLast(-1).timestamp < THRESHOLD:**
Compares the time gap between two consecutive trades with a threshold.

It is also crucial to identify whether the trader fails to utilise safety mechanisms, as this may indicate overconfidence. For instance, neglecting to implement a stop-loss mechanism, which is an essential component of effective risk management, could suggest an assumption of invulnerability, where the trader believes that their trades will not result in losses, which is an evident sign of overconfidence:

Source code 2.8

Pseudocode example overconfidence 3

```
\\ Pseudocode
if !trade.has_stop_loss
    return "Warning: you are not using safety mechanisms, which may be a
           sign of overconfidence"
```

Where:

- **!trade.has_stop_loss:** Checks if the trader has a stop-loss set up for the trade.

2.5 Framing

The framing effect is a cognitive bias where decisions are influenced by the context or wording of information, rather than the underlying facts themselves (Tversky & Kahneman, 1981). In essence, the same probability can be perceived differently, depending on the wording or context. For example, the same probability can appear more attractive when presented as a "70% chance of gain" rather than a "30% chance of loss", even though the numerical risk is identical.

In financial decision-making, especially trading, framing can shape risk preferences, with positive frames encouraging risk aversion and negative frames encouraging risk-seeking (Kahneman, 2011; Tversky & Kahneman, 1981). Framing is not limited to the wording of messages: it also extends to how information is visually presented. As an example, displaying negative

Profit and Loss (P&L) values in red colour may reinforce the perception of a loss, potentially triggering stronger emotional reactions or risk-seeking responses, whereas a more neutral or absent display can lead to different decision outcomes.

The concept of framing lives deeply in a trader's mind, making algorithmic detection challenging. Therefore, mitigation efforts should focus on the design of the user interface, ensuring that data, notifications, and alerts provided to the trader are neutrally worded and avoid emotional language. By standardising how information is presented and avoiding sensational or leading phrasing, the application can minimise the unintentional impact of framing on user decisions.

2.6 Endowment Effect

The endowment effect refers to the tendency for individuals to assign higher value to items they own, compared to identical items they do not own (Kahneman et al., 1990). In Kahneman's studies, individuals given an item (coffee mug) were willing to sell it at a higher price than others offered to pay, implying the fact that the ownership itself increased the item's value. This bias is well-documented in experimental economics: investors may hold on to assets for too long simply because they own them, even in the face of declining value. In crypto trading, this can manifest as the "bag-holding" effect, which refers to refusing to sell depreciated tokens due to a sense of loss aversion and ownership (CoinMarketCap, n.d.). For instance, an investor might hold a token that has lost most of its value since buying, simply because it feels like giving up something valuable. This leads to underperforming and stagnating portfolios. It is worth noting that, in practice, the detection strategy for the endowment effect overlaps with that used for loss aversion, since both biases often lead to similar trading behaviours: holding losing positions for extended periods. As described above in the context of loss aversion, the detection algorithm monitors the duration and performance of held assets, and this same approach is also effective for identifying the endowment effect. While the psychological origins of the two biases differ, the resulting behavioural patterns in trading are similar enough that a unified detection method is appropriate.

2.7 Anchoring

Anchoring bias occurs when individuals rely too heavily on an initial value (the "anchor") when making subsequent judgments or decisions (Tversky & Kahneman, 1974). In trading, common anchors include the original purchase price, a previous price peak, or a recent market high. Traders fixate on these reference points, and it "warps" their judgment of value. For example, if a trader buys a coin for \$50,000, this price becomes their anchor. They might insist on selling the coin only when it returns to this price (assuming the price declined), even if market conditions have changed significantly. Anchoring might engage risk-seeking behaviour, resulting in irrational, prolonged holding of losing positions (similar to Endowment Effect

and Loss Aversion). It can also drive traders to utilise martingale strategies, progressively increasing their position size after losses in the hope of returning to the anchor value. While the anchoring bias provides the psychological motivation (fixation on a reference point), the martingale strategy itself is a form of risk-seeking behaviour, as it involves taking on greater risk to recover prior losses.

The detection strategies for anchoring overlap with those for loss aversion and the endowment effect, as these biases often produce similar trading behaviours, particularly unwillingness to realise losses and persistence in holding underperforming assets. Monitoring the duration and performance of held assets, as described above, can be an effective method for identifying anchoring-driven behaviour.

2.8 Mitigation Strategies

While the identification of behavioural biases is an important step, effective mitigation also requires strategies that help traders avoid acting on these biases. One possible mitigation strategy is the use of cooldowns: brief, forced breaks that prompt users to reflect before making potentially harmful decisions. Such interventions may encourage a shift from fast, automatic responses of System 1 to more deliberate, analytical thinking of System 2 (Kahneman, 2011). Research in behavioural economics further supports the idea that prompts to slow down, reflect, or reconsider, such as nudges or cooling-off periods. Such efforts can interrupt impulsive actions and help individuals make more rational decisions (Ariely, 2009; Loewenstein, 2005; R. Thaler & Sunstein, 2009).

Additionally, when the risk of an ongoing bias is high, cooldowns could be reinforced with targeted questions that encourage deeper reflection. For instance, if signs of overtrading are detected and a user attempts to open new positions, the system could prompt questions like "for how many coins are you opening positions?" or "what is the minimum 24-hour price of this asset?", requiring the user to answer these questions. This would prompt them to reexamine market data or review their recent actions, engaging more analytical thinking of System 2.

In certain situations, where risk indicators suggest almost certain probability of biased behaviour, the system could go further and temporarily block the user from performing specific operations. This type of intervention, though more restrictive, is intended to protect users from severe financial harm and to provide additional protection in high-risk scenarios. However, it is important to consider that overly restrictive measures may have unintended consequences. Research in psychology and behavioural economics suggests that when individuals perceive their freedom of choice is threatened, they may experience psychological reactance and attempt to restore their autonomy by seeking ways to bypass or avoid such restrictions (Mühlberger et al., 2015). Additionally, empirical studies indicate that users subjected to restrictive nudges often feel less satisfied with their choices and may view such interventions as manipulative (Michaelsen et al., 2021). Therefore, finding the right balance between

effective guidance and user autonomy is crucial when designing risk mitigation strategies.

2.9 Summary

The biases and heuristics explored in this chapter demonstrate how cognitive distortions can lead to irrational decision-making in crypto trading. In order to mitigate the risks associated with these biases, an automated risk management module can be employed to detect and address these behavioural patterns in real-time.

Table 2.2 summarises the main features of a proposed automated risk management module that leverages behavioural bias detection to improve trading decision-making. Each feature is designed to automatically identify and prevent specific behavioural biases in trading activity. The system observes key parameters of market data and activity metrics, and detects deviations from established baselines or thresholds.

Feature	Description	Key Parameters
Overconfidence detection	Detects excessive trading due to inflated self-perception.	Daily trade count, time between trades, trade volume, stop-loss status
Loss Aversion - Risk-Seeking detection	Identifies holding losing positions too long to avoid realising a loss.	Unrealised P&L, holding time
Loss Aversion - Risk-Averse detection	Detects premature profit-taking behaviour, closing winning positions too early.	Realised P&L, number of closed positions
Overweighting Small Probabilities detection	Detects investment in low-liquidity, high-risk assets, often driven by rare-event speculation.	Asset liquidity, trading volume
Availability Heuristic detection	Identifies decisions influenced by recent news or short-term fluctuations, leading to emotional reactions.	Asset volatility, price change, news frequency
Irrational Decision-Making Prevention	Integrates risk signals from multiple biases to detect and intervene in potentially irrational trading actions.	Composite risk score, intervention triggers

Table 2.2

Risk-related Requirements

This automated approach reduces cognitive load by offering objective, data-driven feedback, helping traders make more informed decisions and mitigate the impact of biases in the volatile crypto market. The features outlined in the table lay the groundwork for a practical system to

detect and address these biases in real-time. The next chapter will explore the design, architecture, and integration of this risk management module, demonstrating how this framework can be turned into a functional tool for traders.

3. Design and Implementation of the Application

This chapter details the design and implementation of a risk-mitigating system and its supporting infrastructure within a trading platform. The system’s primary goal is to detect and mitigate cognitive biases and risky behaviours in real-time, enabling more rational decision-making in a volatile crypto market. The whole system architecture consists of four primary components:

1. **Trade Engine:** Developed by **my supervisor** in Rust, this component handles the execution of half-automated trading strategies. It is responsible for managing the lifecycle of trades and providing user-related market data.
2. **Backend for Frontend (BFF):** This service is an intermediary between a trader and other system components. It manages user sessions, serves data from other components and handles data persistence.
3. **Risk Detection module:** The core of the platform’s behavioural risk mitigation functionality, this module is responsible for monitoring user trading activities and evaluating risk scores in real-time.
4. **Mobile Application:** The user interface for the system, the mobile application allows traders to interact with the platform on their mobile devices. Users can create and manage strategies through the app, view risk evaluation results, and receive notifications on high-risk activities.

To guide the development of the system, a set of requirements was established based on discussions with my supervisor and aligned with industry standards for building secure, scalable, and high-performance applications. These requirements are not only focused on the functional capabilities of the system but also on non-functional aspects such as performance or security, all of which are essential for implementing such a system. These requirements follow common principles in software engineering, ensuring that the system can handle high volumes of data, be secure, and provide a user-friendly experience (Bass et al., 2012).

The following table summarises the general requirements for the system, providing a comprehensive overview of the design, functional, and security considerations that guided the development of each system component.

Category	Requirement	Description	Key Considerations
Design Choice	User Interface Design	Responsive, intuitive interface	Adaptable and intuitive layout.
Performance	Scalable, high-throughput data processing	Ensures the system can handle high data volumes in real-time.	High-throughput, distributed messaging systems and event-driven architecture.
Functional	User Authentication	Users can register and log in securely via the mobile app.	Secure password storage, session handling.
Functional	Push Notifications	Real-time notifications on risk evaluations.	Send notifications based on risk score.
Functional	Risk Management	Users can set custom limits for risk evaluation.	Functionality to set custom thresholds.
Security	Access Control	Ensure proper user role management for access to sensitive data.	Role-Based Access Control (RBAC) for different levels of access.
Security	Password Hashing	Hash passwords securely before storing them in the database.	Use of secure hashing algorithms
Security	API Key Encryption	Users can link exchange accounts by providing API keys. API keys should be stored securely.	Secure and robust encryption
Security	Secure API Communication	Ensure secure communication between internal services.	Use of HTTPS for all communication, secure authorisation framework for internal services

Table 3.1*General requirements for the System*

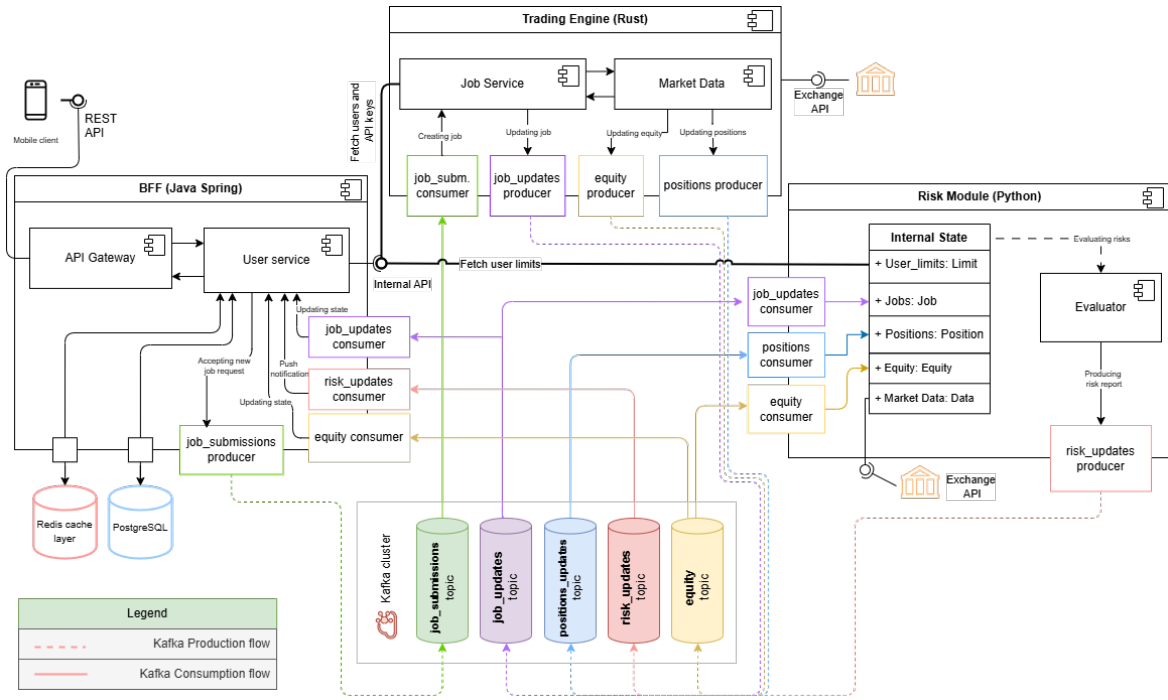
3.1 Overall System Architecture

The system architecture follows a microservices-oriented design, where a dedicated component handles each primary concern. Such components are designed to focus on a single responsibility, allowing easier maintenance and wider scalability. Main components - **Trade engine**, **Backend for Frontend** and **Risk Detection module** - are decoupled but inte-

grated through defined interfaces and messaging channels, primarily using **Apache Kafka** for communication. This approach leverages an event-driven architecture that is well-suited for applications requiring real-time data processing, such as a trading platform. In an event-driven system, services are triggered by a specific event, such as a change in the status of a trade or a risk evaluation being completed. This allows the whole platform to process tasks asynchronously and independently, ensuring the services remain decoupled and scale without interfering with other system parts. The event-driven architecture, combined with a microservice approach, is a good fit for the platform for several reasons (Newman, 2015):

- **Real-Time Processing:** Events in the system, such as trade updates, risk evaluations, or changes in positions, need to be processed in real time to ensure timely responses and notifications to users. An event-driven approach ensures that each component can react to events as they happen, without delays caused by tightly coupled systems.
- **Scalability:** With event-driven architecture, each service can be scaled independently based on demand. For instance, if the Risk Detection Module needs to process more data due to increased user activity, additional module instances can be deployed to handle the increased load, without requiring changes to other system parts.
- **Fault Tolerance:** The system gains greater fault tolerance by decoupling services. If one service encounters an issue, it does not bring down the entire system. However, the service's functionality will be unavailable. Events can be queued and processed later, allowing other services to continue operating. This is crucial in a trading environment, where uptime and reliability are critical.
- **Developmental Flexibility:** An event-driven architecture facilitates the independent evolution of various services. For instance, introducing a new risk evaluation algorithm can lead to an update of the Risk Detection Module without affecting the Trade Engine or the BFF Service, thereby minimising the need for coordinated updates across different services.

To provide a clear view of how the components are connected and interact, Figure 3.1 below represents a high-level architecture diagram of the system.

**Figure 3.1**

High-level UML-based architecture diagram of the system

3.1.1 Data Flow in System and Kafka's role within it

As mentioned above, the system's services operate independently, focusing on a specific task, which makes it essential for them to have a reliable and efficient mechanism for communication. Since the system is built around an event-driven model, where services trigger specific actions based on events (e.g., trade updates, risk evaluations), the underlying communication system must effectively handle these events as messages, support scalability and be fault-tolerant. In this context, **Apache Kafka** was chosen as the central message broker due to several reasons, which align well with the system's requirements (Kreps, 2011):

- **High throughput and scalability:** Kafka can handle very high message volumes with low latency, making it ideal for a real-time computing platform.
- **Fault tolerance and durability:** Kafka provides strong guarantees for message delivery, replication, and persistence, which is crucial in financial systems where data loss or downtime cannot be tolerated.
- **Decoupling of services:** Kafka allows independent services to communicate asynchronously, promoting modularity and ease of maintenance.

Kafka Topic	Description	Publisher	Listener
job_updates	Topic used to publish job events (created/paused/stopped/finished).	Trade Engine	BFF, Risk module
job_submissions	Topic used to accept new user jobs from the mobile app into the system.	BFF Service	Trade Engine
positions_updates	Topic used to gather actual user positions data from exchange (Bybit, Binance).	Trade Engine	BFF, Risk Module
equity	Topic used to gather equity of exchange account (bybit, binance).	Trade Engine	Risk module, BFF
risk_updates	Topic used to publish risk evaluation results.	Risk Module	BFF

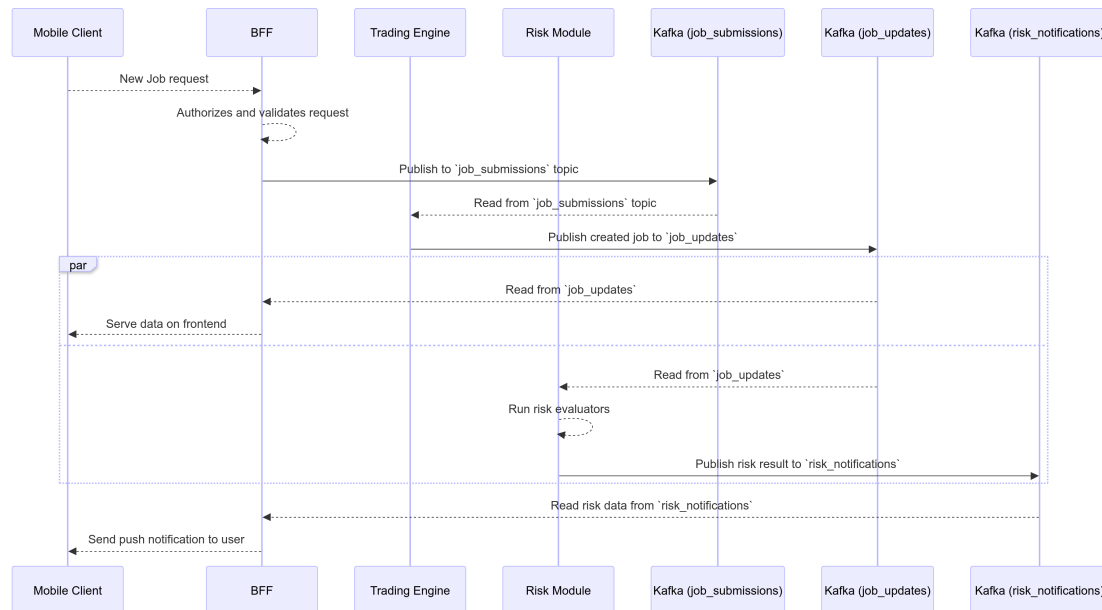
Table 3.2*Kafka topics and their roles in the system***Figure 3.2***Sequence diagram of job creation process*

Figure 3.2 illustrates the interactions between individual components during a typical job creation process: the mobile app communicates via REST API with the BFF, which authorizes the request and passes it further to the trade engine through a specially assigned Kafka topic, which is listed in Table 3.2. The Trade engine accepts the job request and produces updates to the dedicated Kafka topic, which is being listened to by the Risk module. The risk module receives job and position updates, triggering risk evaluation, the results of which are pushed into the corresponding Kafka topic. If the risk level is significant, BFF will send

a push notification to the mobile app, informing the user.

By employing Kafka as the central message broker, the platform ensures that high-volume data streams (e.g. job events, risk evaluation results) can be handled with low latency and high throughput. Each component, besides BFF, can scale horizontally. Multiple instances of the Risk module can be run simultaneously in a Kafka consumer group to process partitions of the data stream in parallel, each instance processing its dedicated partition, which is allocated to a limited range of users (e.g. partition 1: users 1-1000, partition 2: 1001-2000, etc.)

From an architectural perspective, each service adheres to the single-responsibility principle - the Trade engine focuses on trade execution logic, BFF focuses on session management and data serving, the Risk module focuses on risk evaluation, and the mobile app focuses on data presentation and user interaction.

3.2 Trade Engine

The Trade engine is a standalone service written by my supervisor in Rust. While I did not contribute to the development of the service, it is important to briefly describe the design of the service, since other services were designed to work alongside. The engine supports two types of half-automated strategies, which are referred to as jobs:

- **Dollar-Cost Averaging (DCA):** An investment strategy where a fixed amount of capital is invested at regular intervals to purchase a target asset. Parameters include the assets, investment amount, strategy duration and number of steps, price discount, and an optional `force` flag. When enabled, the `force` option executes the purchase immediately at the current market price, ignoring the price discount condition.
- **Liquidation (LIQ):** A symmetrical strategy focused on closing positions rather than opening them. The same parameters apply: the asset, amount to sell, time horizon with number of steps, price discount, and the `force` option. In LIQ, the `force` flag triggers immediate position exit at the market price, bypassing any discount logic.

The service uses three Kafka topics: `job_events` to publish any updates on active jobs, `job_submissions` to accept new job requests from the users and `positions_updates` to publish data about user's positions, including but not limited to positions size, unrealised and realised P&L, position side.

3.3 Backend for Frontend

The Backend for Frontend (BFF) service is responsible for managing user sessions and orchestrating data between the frontend and internal servers. Implemented with Spring Boot,

it handles user authentication, session management, persistent data storage, and communication between the Trade engine, Risk module and mobile application.

3.3.1 User Sessions

The user authentication process is implemented using Spring Security and JSON Web Token (JWT) with HS256 as a signature algorithm, which provides a strong protection layer against common security threats (Rana et al., 2024). When a user logs into the app (with credentials or Discord OAuth protocol), the BFF authenticates the user and issues a JWT signed with the server's secret key. This token contains claims such as the user's identity (username, email) and the user's roles, which define permissions. The token is then used as a header for every HTTP request, allowing the server to verify the user's identity and authorise access to protected resources. Each request is validated by checking the token's integrity and expiry, ensuring secure and authorised interactions between the frontend and the service.

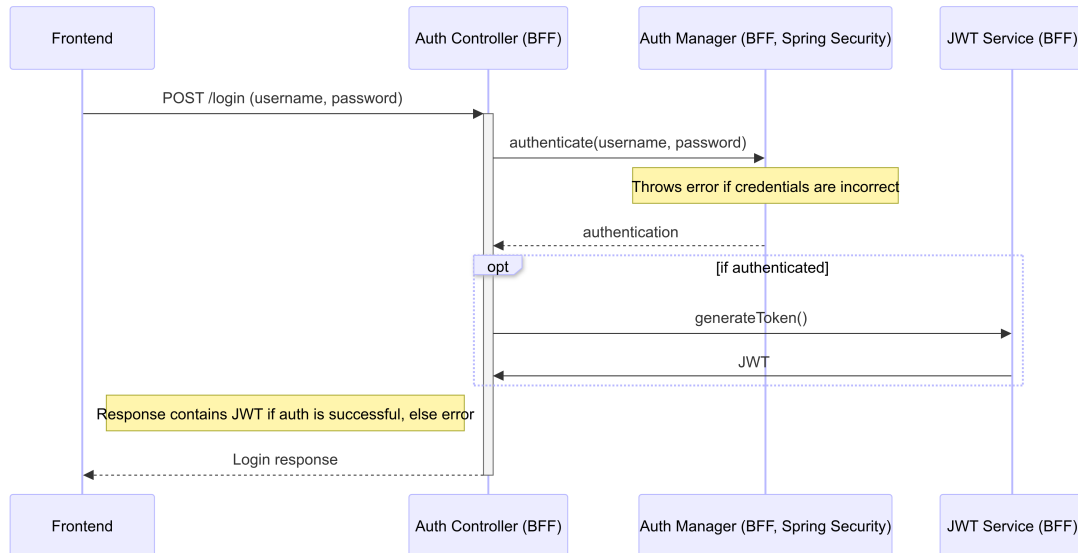


Figure 3.3

User authentication flow

3.3.2 Data Persistence

The BFF service uses a relational database to persist core data, such as user accounts, user third-party accounts such as Discord or exchange platforms (e.g., Bybit, Binance), jobs, notifications and other account-specific data. The relationships among these entities are illustrated in Figure 3.4, which presents an overview of the persistence layer's class structure.

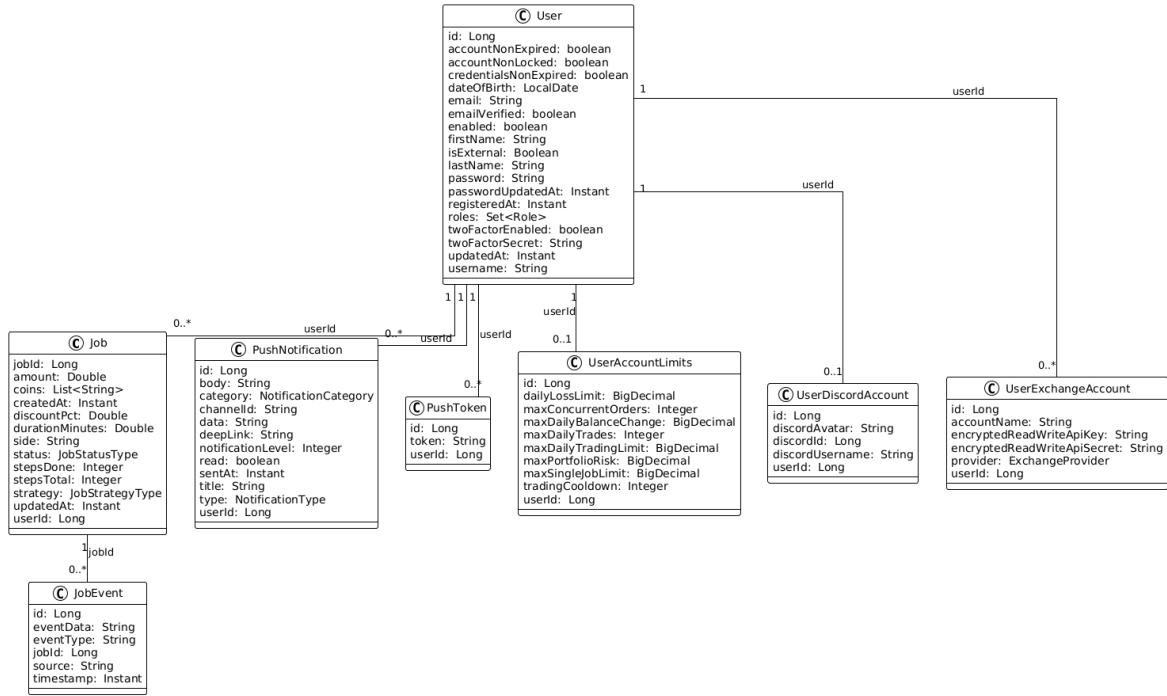


Figure 3.4

UML Class Diagram of Persistence Layer

3.3.3 Push Notifications Service

When certain events occur, for example, a scheduled DCA job is finished or the Risk module flags a high-risk pattern, the BFF notifies the user by using push notifications. Service uses Firebase Cloud Messaging to ensure both Android and iOS are supported. The BFF maintains mapping of users to device tokens (obtained when a user authenticates via the mobile app).

3.3.4 API Design

The BFF exposes a set of REST API endpoints to the mobile application and internal system services. For enhanced security, endpoints serving highly sensitive data (such as API keys) are only accessible from trusted, internal network addresses (e.g., 127.0.0.1). For instance, when the **Trade engine** needs to retrieve a user’s exchange API keys, it must call a dedicated BFF endpoint that is restricted to a local, private network, ensuring regular users or external parties cannot access this sensitive information. While this approach provides a basic layer of protection, modern security best practices recommend not relying solely on network location for trust (Rose et al., 2020), which will be addressed in Conclusions.

In addition to traditional RESTful APIs, the BFF also provides real-time data updates to clients via Server-Side Events (SSE). The SSE endpoint delivers a continuous stream of updates, including market data, user positions, equity state, job updates, and risk reports

directly to the mobile application. This design choice ensures that users receive the latest account and market data without needing to repeatedly poll the server (MDN, 2025). The use of SSE improves the responsiveness and user experience, especially for time-sensitive trading and risk events.

A full specification of the available endpoints is listed in Appendix H.

3.3.5 Additional Security Measures

The BFF service implements multiple layers of security to protect user data and maintain system integrity:

- **Password Hashing:** All user passwords are hashed using BCrypt cryptographic algorithm, plaintext credentials are never persisted. This approach is widely recommended for storing user passwords securely (Grassi et al., 2017).
- **API Key Encryption:** Exchange API keys are encrypted in the database using AES-GCM (128-bit), in line with current NIST recommendations for protecting sensitive data (Barker, 2020).
- **One-Time Password (OTP):** For operations such as login (if enabled) and password change, an additional OTP verification is required. The use of OTPs is a best practice for multi-factor authentication (Grassi et al., 2017).

3.4 Risk module

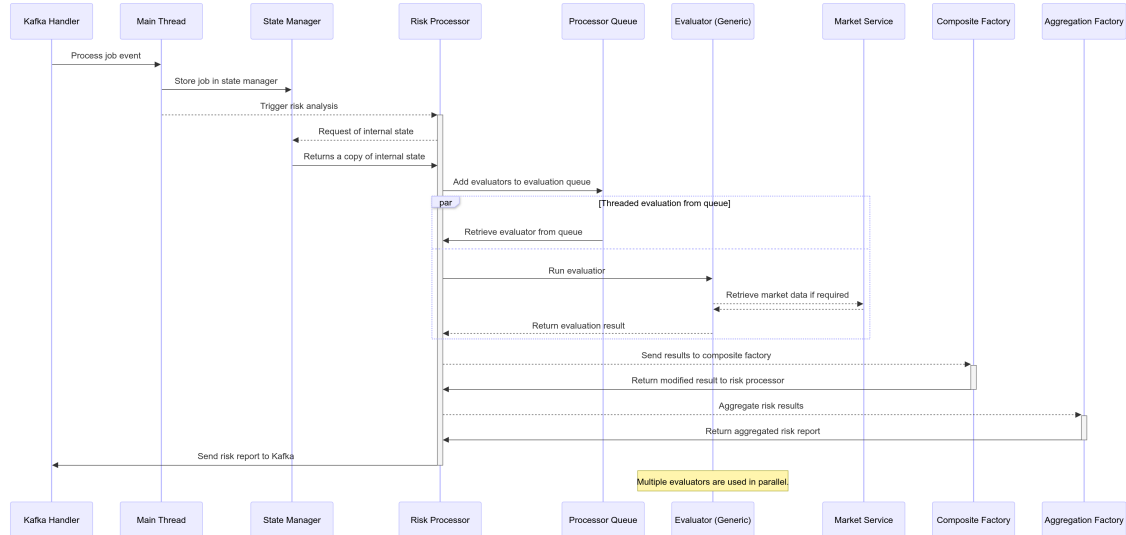
The Python-based Risk module is the centre of the platform’s risk-mitigating feature. Its purpose is to continuously monitor user trading behaviour and detect bias-related patterns, as described in the previous chapter.

3.4.1 Module Architecture

The risk detection module is structured as an event stream processor that consumes events, builds its internal state, and produces risk evaluation. The figure 3.5 can be referenced to visualise the key components within the module, which include:

- **Kafka Handlers:** Three consumers subscribe to `job_updates`, `positions_updates`, and `equity` topics. Events are deserialised from JSON and added to the module’s internal state.
- **Risk Processor:** Coordinates evaluator execution. Uses a `ThreadPoolExecutor` and request queue to ensure concurrent, non-blocking evaluations under high throughput.
- **Composition Factory:** Matches atomic patterns against predefined behavioural rules to generate composite patterns with confidence scores.

- **Market Data Service:** Provides real-time market data (e.g., candlesticks, orderbook) via Bybit API for risk evaluation.
- **Aggregation Factory:** Aggregates pattern detections into a final risk report, computing category-level confidence.
- **Kafka Producer:** Publishes structured risk reports to an output Kafka topic for further processing within other services.

**Figure 3.5**

Sequence diagram of Risk Module processing

To improve interpretability for users, all biases discussed in Chapter 2 are mapped to one of three high-level categories:

Table 3.3

Behavioural Risk Categories

Category	Description
Overconfidence	Reflects excessive or aggressive trading behaviour, such as overtrading, ignoring limits, or skipping cooldowns.
FOMO	Indicates emotional reactivity to hype or volatility, e.g., trading illiquid assets or assets with increased interest over the news (Availability Heuristics).
Loss Behaviour	Includes both risk-averse (cutting profits too early) and risk-seeking (holding losing positions too long) behaviour.

3.4.2 Supported Metrics

The module currently supports the calculation of various market and trading metrics, which are used to inform pattern detection. These metrics are derived from the Chapter 2.

Market Volatility

Volatility is computed using historical price data from Bybit. The system calculates the standard deviation of returns and annualises the result to normalise across timeframes.

Source code 3.1

Volatility calculation from close prices

```
returns = np.diff(closes) / closes[:-1]
std_dev = np.std(returns)
annualized_vol = std_dev * np.sqrt(252 * 24 * 60 / interval_minutes)
```

Liquidity Metrics

The system measures market liquidity using real-time orderbook data. Two key metrics are derived: the spread between the best bid and ask prices, and the total order volume within a 1% price range around the mid-price. These figures reflect transaction costs and market depth, respectively.

These metrics are used to flag positions opened in illiquid or high-spread markets. Spreads above 2% or depth below \$100 000 are treated as potential risk factors.

Hype Score

While the Availability Heuristics discussion in section 2.3 suggested that spikes in news frequency could indicate potential hype-driven behaviour, practical limitations of most news APIs (e.g., strict rate limits or limited crypto coverage) made this approach problematic. Instead, the system uses Google Trends as a proxy for social interest.

The `pytrends` Python library is used to retrieve historical search interest for crypto assets. Asset symbols (e.g., "BTC") are mapped to full names (e.g., "Bitcoin") for better search resolution.

The hype score is calculated by comparing the most recent search volume against a multi-day historical average. This deviation is then scaled into a normalised score between 0.0 and 1.0:

Source code 3.2*Simplified hype score calculation*

```
deviation = (current - average) / average
hype_score = max(0.0, min(1.0, base_score + deviation))
```

The score reflects how much current public attention deviates from recent norms. Higher values indicate elevated attention to the asset, which can be a sign of ongoing Availability Heuristics bias.

User Limit Violations

Rather than relying on hardcoded thresholds, the system allows users to configure their own trading limits. These parameters are used as baselines to evaluate whether recent actions deviate significantly from self-defined constraints. This flexibility supports personalised risk assessment aligned with individual trading preferences and capacities.

Limit Parameter	Purpose
Max Single Job Limit	Maximum allowed amount (in USDT) for a single job
Max Daily Trading Limit	Maximum total trading volume (in USDT) permitted
Max Portfolio Risk	Maximum percentage of the portfolio that can be allocated to a single position
Max Concurrent Jobs	Maximum number of jobs allowed to be running at the same time
Max Daily Trades	Maximum number of trades allowed within a 24-hour period
Trading Cooldown	Minimum required interval (in minutes) between consecutive jobs
Daily Loss Limit	Maximum daily unrealised loss (in USDT)
Max Daily Balance Change	Maximum allowable percentage change in total balance within a day

Table 3.4*User-Defined Limits Used in Risk Evaluation***Position-Based Metrics**

Several indicators are derived from position data, using historical position updates:

- Exiting profitable positions too early (P&L < 0.05)

- Holding losing trades too long (≥ 7 days)
- Unrealized P&L of position ($\leq 10\%$)

All features mentioned above allow the module to observe and evaluate the full range of behavioural indicators discussed in Chapter 2. The computed metrics serve as the basis for detecting risk-related patterns and ensure that all critical inputs outlined in the system specification (see Table 2.2) are actively monitored and evaluated.

3.4.3 Risk Detecting Functionality

As discussed in Chapter 2, behavioural biases can not be identified by a single isolated action, but rather from a combination of simpler deviations from optimal behaviour, which this system refers to as *atomic* patterns. These atomic patterns are simple, context-free risk signals. On their own, each provides limited information, but when combined, they can reveal more complex behavioural tendencies.

Each atomic pattern is associated with a severity score that represents the degree of deviation. Severity is calculated dynamically based on a *violation ratio*, defined as the discrepancy between the observed value and a baseline or configured threshold. For example, if a user performs 15 jobs in a day where the maximum allowed is 10, the violation ratio is $R = \frac{15}{10} = 1.5$.

To translate this into a normalized severity score $S \in [0.0, 1.0]$, a logarithmic scaling function is used:

$$S = \min \left(1.0, \max \left(0.0, \frac{\log_2(R)}{\log_2(V_{\max})} \right) \right)$$

where:

- R is the observed-to-expected ratio
- $V_{\max}$ is the maximum violation ratio treated as a full-severity breach (e.g., 2.0)

This transformation ensures that minor deviations are scored moderately while still allowing extreme behaviours to saturate the scale.

In some cases, lower values are considered more severe. For example, opening a new job too early after a previous one may violate the cooldown. Such cases are handled using an inverted version of the same formula:

$$S = \min \left(1.0, \max \left(0.0, \frac{\log_2 \left(\frac{V_{\max}}{R} \right)}{\log_2(V_{\max})} \right) \right)$$

In addition to severity, atomic patterns carry *category weights* that map them to behavioural risk categories such as *FOMO*, *Overconfidence*, or *Loss Behaviour*. While these weights can be manually specified, doing so requires domain knowledge and justification. Therefore, when weights are not explicitly set, the system distributes relevance equally across all defined categories, assuming a single atomic pattern can not be treated as a sign of specific bias.

While atomic pattern severity scores are always used for computing the confidence of composite patterns, their category weights are typically overridden. Atomic weights only contribute directly to the final behavioural evaluation if no composite patterns are matched, and this fallback mechanism is explained in detail in the aggregation section.

Once atomic patterns are collected and scored, the system attempts to match them against predefined *composite* pattern rules. Each composite pattern represents a known bias behaviour, such as overconfidence or loss-related behaviour. The current system supports the following composite patterns:

- **Overtrading** — Triggered by both high daily trade count and total volume. Indicates potential overconfidence.
- **Trading Burst** — Based on frequent cooldown violations, large trades, and use of forced job parameters. Suggests impulsive or overconfident behaviour.
- **Risk-Seeking Behaviour** — Detected when a user holds a losing position for an extended period. Reflects loss aversion bias.
- **Premature Profit-Taking** — Triggered by multiple early exits from small profitable positions. Indicates risk-averse decision-making.
- **Speculative Allocation** — Identifies large trades in illiquid markets. Suggests over-weighting of small probabilities.
- **Availability Heuristic** — Detected when trading occurs in assets with high volatility and social interest. Suggests asset allocation is influenced by news, assuming FOMO behaviour.

The goal of this process is to move beyond isolated deviations and identify clusters of related atomic patterns that together indicate bias-affected behaviour.

The matching logic is managed by composition rules, which define the minimum atomic patterns required, whether the order of patterns matters, and the timeframe in which the patterns must co-occur. For example, a rule might specify that a cooldown, single job amount violations and a job with "force" parameter must all occur within a 60-minute window to indicate a potential burst of trading. These rule definitions allow the system to express both strict (sequence-based) and flexible (window-based) behavioural patterns.

Source code 3.3

Sample composite pattern rule for detecting trading burst.

```
CompositePatternRule(  
    rule_id="trading_burst",
```

```

pattern_requirements={
    "limit_cooldown": "1+",
    "limit_single_job_amount": "1+",
    "limit_force_job": "0+"
},
category=RiskCategory.OVERCONFIDENCE,
time_window_minutes=60,
sequence_matters=False,
message="Burst of trading activity detected"
)

```

Where:

- **pattern_requirements** Required atomic patterns and their number.
- **category** Main pattern category.
- **sequence_matters** Whether the order of patterns matters.
- **time_window_minutes** A time window during which all patterns must occur.

The matching engine first filters atomic patterns by recency and then groups them by pattern type. It validates that all required patterns exist in sufficient quantity. If the rule allows unordered matching, the engine searches for any group of patterns that satisfy all constraints within the time window. If the sequence is important, the engine verifies the order of occurrence.

If a valid match is found, a composite pattern is instantiated. This pattern contains an associated behavioural risk category and the total confidence score. This score $C \in [0.0, 1.0]$ is calculated as the average of the severities of the matched atomic patterns:

$$C = \frac{1}{n} \sum_{i=1}^n S_i$$

where S_i is the severity of the i -th atomic pattern, and n is the total number of matched components. This averaging approach balances local intensity with behavioural consistency, ensuring that the pattern reflects not just a single anomaly but a sustained deviation from rational behaviour. Matched atomic patterns are flagged as "consumed", preventing them from being reused in other composite patterns unless explicitly allowed.

Once a composite match is made, its score becomes the dominant input in the user's risk evaluation.

This layered approach, which starts with simple behavioural signals and builds up to structured patterns, allows the detection system to balance two goals: warning users early about possible risks and only flagging behaviour when there is strong evidence of a bias. In addition, such a design choice provides flexibility, so new atomic or composite patterns or updates to

existing ones can be integrated with relative ease as new features are added.

Aggregation and Risk Scoring

Once atomic and composite patterns have been identified, the system aggregates them into a risk report. This report captures the most relevant behavioural signal at the current moment by evaluating all available patterns and their assigned weights.

The aggregation process prioritises composite patterns. If any composite patterns are present, the system uses only these to compute final confidence scores, assuming they reflect more structured behavioural risks. In the absence of composite matches, unconsumed atomic patterns contribute to the overall risk estimation.

For each behavioural category (e.g., *Overconfidence*, *FOMO*, *Loss Behaviour*), confidence is calculated using a weighted average:

$$C_{\text{category}} = \frac{1}{n} \sum_{i=1}^n S_i \cdot w_i$$

where S_i is the severity (or composite confidence) of the i -th pattern, and w_i is its corresponding category weight. For atomic patterns, a discount factor of 0.5 is applied to reduce their influence when composites are not present:

$$w_i^{\text{effective}} = w_i \cdot 0.5$$

The final risk category is determined by selecting the category with the highest aggregated confidence. This top category is then mapped to a discrete risk level:

- **None** if $C < 0.01$
- **Low** if $0.01 \leq C < 0.5$
- **Medium** if $0.5 \leq C < 0.7$
- **High** if $0.7 \leq C < 0.9$
- **Critical** if $C \geq 0.9$

This scoring logic ensures that the final output reflects not just raw pattern detection but a weighted, category-aware confidence. The result is encapsulated in a structured report, which includes:

- The top behavioural risk category and its severity level
- Aggregated scores for all other categories
- A complete list of atomic and composite patterns observed

This report becomes the primary output for behavioural feedback, used by both the mobile application and BFF for alerting.

3.5 Mobile Application

The mobile application is built using the React Native framework, enabling cross-platform deployment for both iOS and Android devices. Its primary role is to serve as the user interface layer, allowing users to interact with the platform, open new jobs, monitor their trading activity, and receive behavioural risk reports.

3.5.1 Application Structure

The app is divided into several main screens, each serving a distinct role, such as job creation or risk overview. The app is developed using the React Native Navigation library, which provides a native-like navigation experience across iOS and Android platforms. The primary parts of the app are accessible via a bottom tab navigator, a widely adopted mobile UX pattern that improves accessibility and consistency across views (Nick Babich, 2022).

Each tab corresponds to a key section of the application:

Home Screen

A minimal dashboard showing general statistics and navigation shortcuts. This screen serves primarily as an entry point and will not be discussed in detail.

Market Section

The Market section offers users a live overview of all supported cryptocurrency assets. It provides market indicators, such as current price, 24-hour volume and other market data. Market data is retrieved from the BFF service using SSE channel, ensuring low-latency updates without requiring constant polling. The full description of data retrieved through SSE can be viewed at subsection 3.3.4.

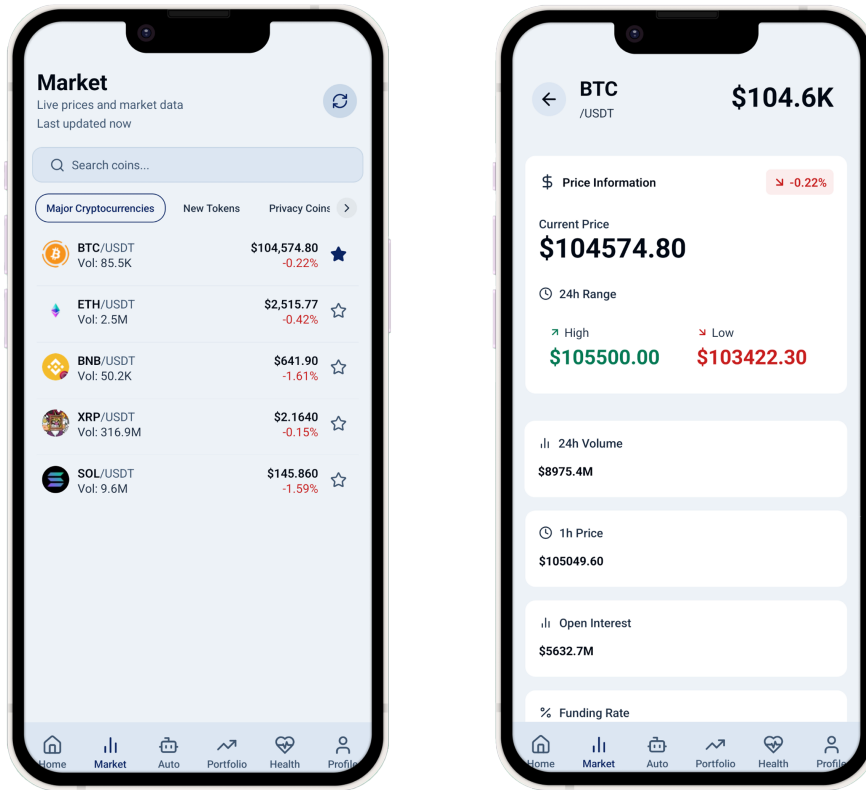


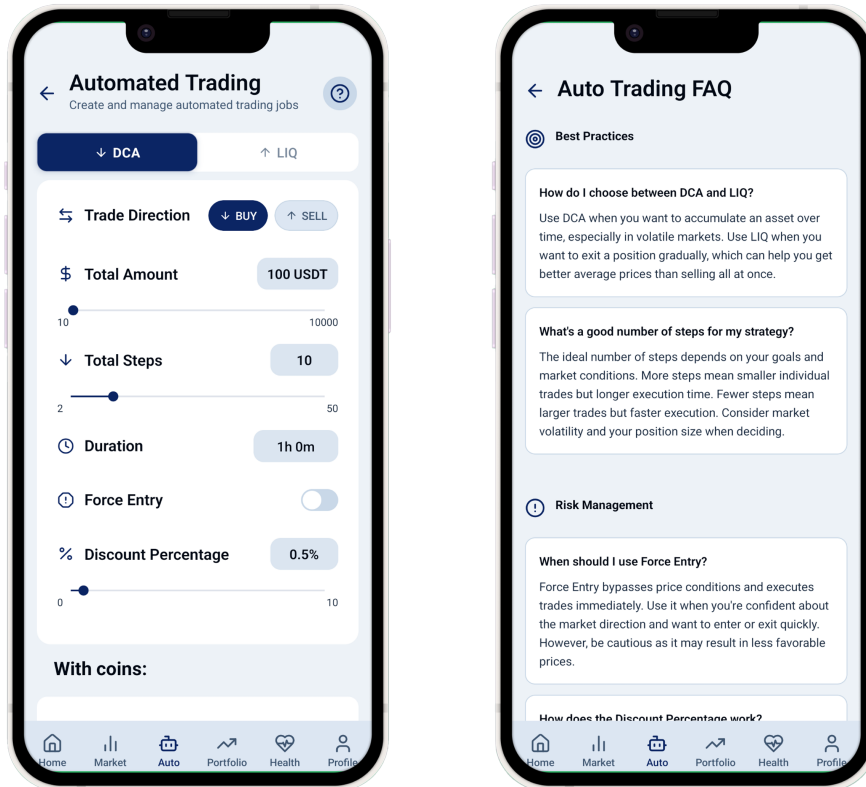
Figure 3.6
Screens from the "Market" section of the Application

Auto Section

The Auto section serves as the interface for configuring and managing automated trading strategies (jobs). Users can create new jobs by selecting coins and configuring key parameters. Submitted jobs are dispatched to the trading service, which handles execution.

In addition to job creation, the section provides an overview of active jobs and a historical log of previously executed strategies.

To assist users, a brief FAQ section is included within the same screen, providing a brief explanation of how jobs work.



(a) Job Creation Section

(b) FAQ section

Figure 3.7

Screens from the "Auto" section of the Application

Portfolio Screen

The Portfolio screen gives users a quick overview of their total account value, available balance, and open trading positions. Users can see real-time updates of their exchange accounts, along with a clear summary of each active position, including entry and current prices, P&L, and trade direction. This layout allows users to monitor their trading performance and manage their positions.

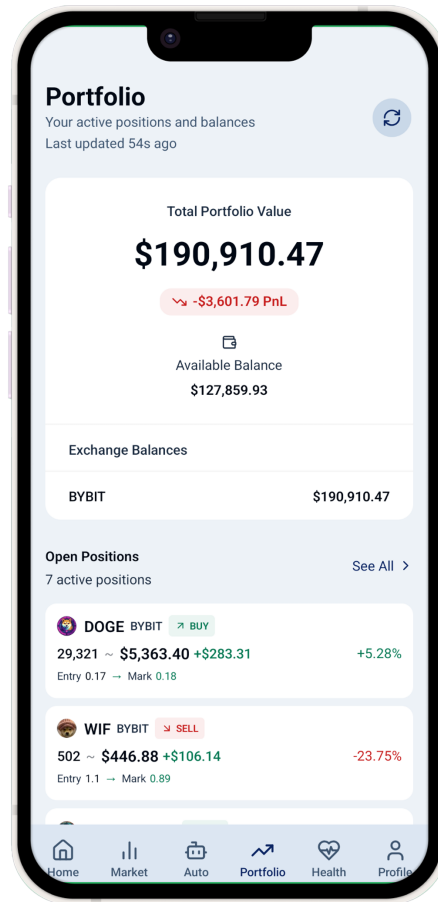


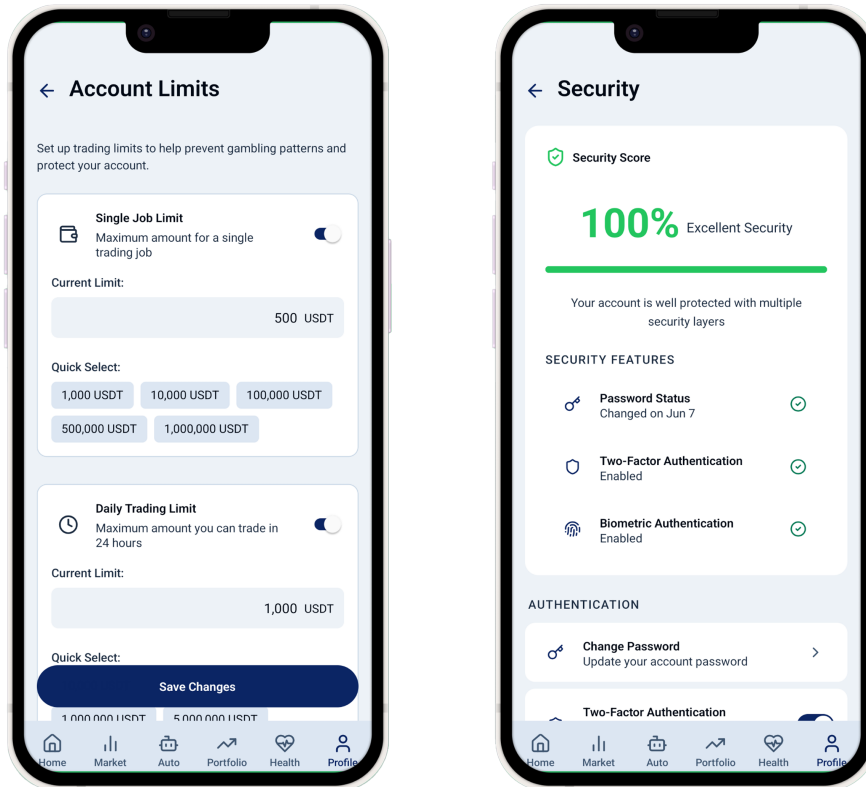
Figure 3.8

Portfolio Section of the Application

Profile and Settings

The Profile section provides users with access to their core account information and management features. Users can connect both exchange accounts (e.g., Bybit and Binance) and social accounts (Discord). This section also includes application-wide settings such as configurable trading limits, security preferences, and options for visual appearance (e.g., dark or light mode).

Particular emphasis was placed on the security settings. The security screen displays the current protection status with a clear “Security Score” and encourages users to strengthen their account security. Security features, such as password change, Two-Factor Authentication (2FA), and biometric authentication are clearly presented with intuitive indicators.

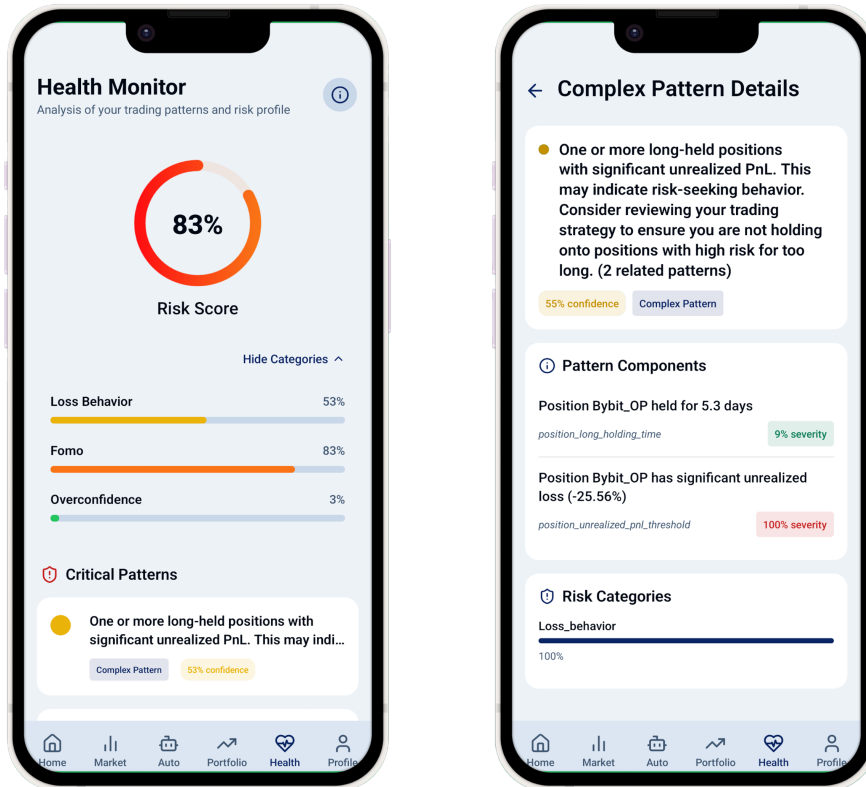
(a) *User Limits Settings Screen*(b) *Security Settings Screen***Figure 3.9**

Screens from the Profile and Settings section of the Application

Health Section

The Health section is a core component of the application, designed to display behavioural risk reports from the Risk module. On this screen, users receive real-time feedback about their trading behaviour. Each report highlights any detected patterns, categorising them by type (such as Overconfidence, FOMO, or Loss Behaviour) and indicating the current risk level.

Users can view all active risk signals, including both atomic and composite patterns, referred to as early warning signs and complex patterns in the app, respectively. The interface allows users to see details for each detected pattern, along with information on how different patterns are related.



(a) Risk Report Screen

(b) Pattern Details Screen

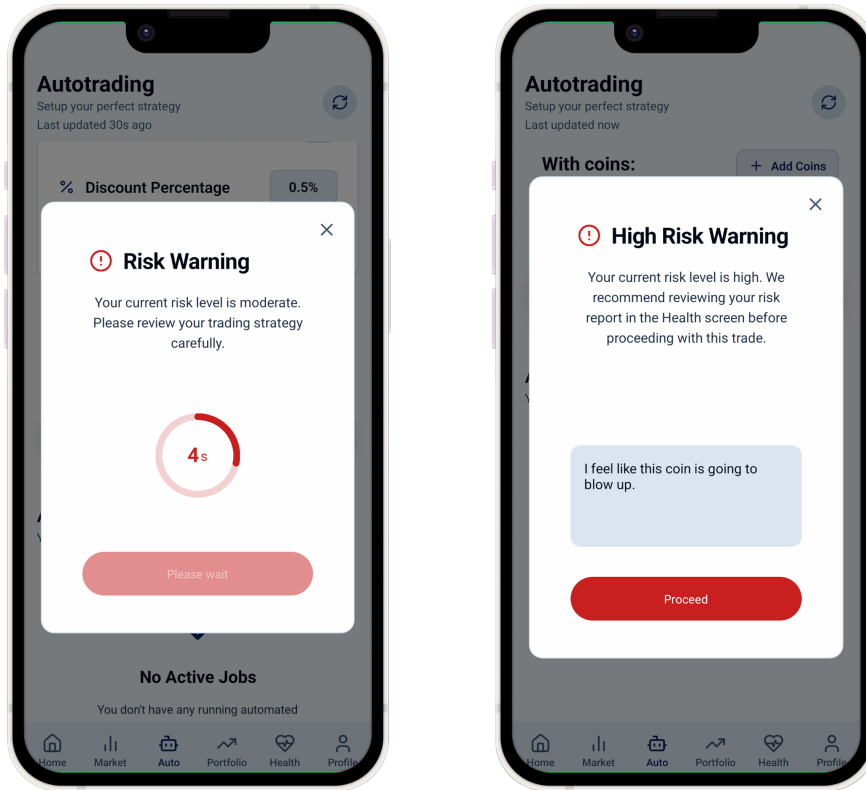
Figure 3.10

Screens from the "Health" section of the application

3.5.2 Risk Prevention Mechanisms

The app also implements risk-mitigation measures to help users avoid impulsive decisions:

- **Medium Risk:** If a user tries to open a new job while at medium risk, the app enforces a 15-second cooldown, shown in Figure 3.11a. This short pause encourages the user to slow down and review their decision before proceeding.
- **High or Critical Risk:** When risk is high or critical, the app additionally requires the user to provide a written justification (see Figure 3.11b) before continuing. This extra step is designed to make users stop and think carefully about their trading strategy.
- **Push Alerts:** The app also sends real-time notifications to alert users when their behavioural risk score is above safe limits.



(a) *Medium Risk Cooldown Modal*

(b) *High Risk Cooldown Modal*

Figure 3.11

Cooldowns modals as they appear for users with risk detected

These cooldown measures are based on research showing that even simple pauses or prompts can help people move from fast, automatic reactions to more thoughtful, deliberate choices, as discussed in Chapter 2.

4. User Testing and Evaluation

This chapter presents an in-depth evaluation of how users interact with the application and perceive its core features, including behavioural bias detection, preventing irrational decision-making, security, and usability. The aim was to assess whether the app functions as intended and delivers meaningful value across a range of user scenarios and experience levels.

4.1 Data Overview and Thematic Coding

Across the three participant interviews, a total of 138 unique quotations were coded. These quotations were grouped into six main themes: five derived directly from the interview structure (*Bias Detection and Impact*, *Security and Trust*, *Design and Usability*, *Beginner Accessibility*, and *Real-World Use*), and an additional sixth theme *Background Information* emerged organically from the data.

Figure 4.1 provides a visual overview of this thematic organisation as produced in Quirkos. Each cluster represents a key area explored in the user interviews, illustrating the breadth and structure of the qualitative data prior to detailed analysis.



Figure 4.1

Canvas view of Quirkos with grouped codes across all interviews.

The following section presents the findings structured according to these core themes and the research questions guiding the evaluation.

4.2 Results of Thematic Analysis

Thematic analysis of the coded interviews revealed several insights across the key identified themes. The following sections present key findings related to each theme, supported by participant quotes.

RQ1: How accurately do users perceive the app’s behavioural bias detections relative to their own decision-making patterns?

Participants generally described the app’s bias detection as consistent with their actual behaviour during trading. Reported examples included detection of specific patterns such as overconfidence, a burst of trading, FOMO and Loss Behaviour patterns. These were recognised and accepted by users as interpretations of their activity:

“It detected the FOMO, because I traded one specific coin [...] and it told me that it’s a popular commodity right now and you shouldn’t trade that.” (P1)

“The last one [pattern of closing positions with little profit] actually surprised me, because it was observing how fast do I close my deals” (P1)

“My expectations were that this app should warn me of risky behaviour and analyse my financial behaviour overall, which it did.” (P2)

“I wanted to try it all. I created a lot of jobs with various coins and settings, and then I received a notification saying that apparently my risk score is high, and then I opened health report, and it was saying that I violated my limits” (P3)

There was general agreement that the system was more precise than participants expected, though a minority expressed scepticism about how well it could understand individual intent, which reveals an important limitation of the app:

“This app could never know why I am making a decision. If it tells me I am affected by a bias, it couldn’t know. It just bases its metrics on some facts.” (P1)

Overall, participants perceived the detection system as mostly accurate and responsive, but limited in its ability to infer internal motivations.

RQ2: How do users perceive and respond to the app’s behavioural bias alerts in the context of decision-making?

Participants generally described the bias detection feature as useful, both in terms of awareness and behaviour. Most highlighted its value in uncovering behavioural tendencies they were not previously aware of, such as impulsivity or overconfidence. Participants noted that the alerts prompted them to reflect on their actions or reconsider decisions they might otherwise have made quickly:

“I didn’t think about that notch that it was a risky behavior” (P1)

“It told me something about myself I didn’t know.” (P1)

“I stopped creating jobs after receiving a warning about overconfidence” (P3)

Some participants also noted the lack of a similar feature in other financial or trading platforms:

“It [Bybit] doesn’t even have this feature, and you’re safe here.” (P1)

“Risk analysis isn’t something I’ve met in other apps” (P2)

While most participants described the risk-related safeguards as useful or reflective, one participant expressed frustration with their design. They felt that certain protective mechanisms were overly restrictive or intrusive, particularly during moments of impulsive decision-making. This participant also showed a generally risk-seeking approach to trading throughout the interview, which may have influenced their reaction to the safeguards built into the application:

“It tried to protect me, but what it did, it just annoyed me, so I clicked buttons faster, hoping it would go through the protection.” (P1)

Despite the criticism, the participant did not reject the underlying intent of the system and acknowledged that the app was responding logically based on their observable behaviour. This illustrates a key design anticipation in behavioural tools: balancing guidance and intervention without triggering user resistance, particularly for those leaning toward irrational decisions.

There was also clear evidence that the feature encouraged participants to think more actively about their own behaviour. Instead of simply slowing down or changing the strategy, participants reported doing additional research or reconsidering their assumptions. In some cases, the alerts led to learning or a shift in how they understood typical trading behaviour:

“With cryptocurrencies, I thought that it was normal to do some weird stuff and buy here, sell here in the course of like a few minutes, but apparently it’s not. So, I guess I even learned something from that” (P1)

“I decided to do research on what I’m buying and why the app is warning me about my behaviour” (P2)

“After the app notified me about FOMO and the overhype surrounding XRP, I chose to research the topic further on the Internet” (P3)

RQ3: How do users evaluate the app’s security measures, and to what degree would they trust it for managing personal and financial data?

Security and trust were important factors for all participants, especially given the app’s integration with external trading platforms and handling of sensitive financial data. Most users expressed general trust in the app’s security model, including features like One-Time Passwords (OTP), Two-Factor Authentication (2FA), and biometric authentication. These were not only accepted but often highlighted as signals that the app “cares about privacy and security” (P3).

“Biometrics feature [...] gave me an understanding that the app really cares about my privacy and security.” (P3)

“After I implemented those, I was sure it was all secure.” (P2)

All three participants stated that they had no reason to suspect a breach of security during their use of the app. An interesting insight was discovered regarding the basis of this trust: rather than evaluating the app based on technical transparency (open-source), participants reported feeling confident in the app’s security because it corresponded with other professional financial tools. The presence of familiar features, such as two-factor authentication or encryption, seemed sufficient to establish credibility.

“it feels like a standard app that I don’t have a problem trusting to” (P1)

“And I felt like I can trust this app because it follows some certain patterns other apps follow.” (P2)

All participants commented on the clarity of data-handling policies:

“There was a notification about how my data would be used. It was very clear.” (P2)

“All the information I provided to the app was stored in an encrypted way.” (P2)

“it does a quite standard job about informing how my data was used” (P1)

While no participant reported direct concerns, participants emphasised the high importance of data protection, mentioning prior breaches in other trading platforms:

“If my financial data were leaked, I wouldn’t be using this app anymore.” (P2)

(In response to a question about what might stop them from using the app)

“It could be something like data leaks because I remember Bybit was data leaked.”
(P3)

In summary, participants generally regarded the app’s security features as sufficient. The presence of standard protective measures, paired with clear communication, appears to have established a reasonable level of trust.

RQ4: How comfortable and intuitive do users find the app’s design and functionality during typical use?

Participants generally responded positively to the app’s interface, describing it as clean and intuitive. They explicitly compared it to other financial apps, stating that this application felt easier to navigate and more streamlined in presentation due to its limited scope compared to other financial apps.

“I think it [the app] looks simple and neat, and everything was in its place, was user-friendly and visually appealing.” (P3)

“I found it actually a bit more comfortable than Binance’s app.” (P1)

“This app, in its design and its interface, was very similar to other financial apps.”
(P2)

Participants appreciated that the app followed familiar patterns from other mobile applications, which likely contributed to a sense of ease in use:

“I think it was very clear to me, and the app’s design follows the patterns of other apps, so it was very intuitive to use.” (P2)

Despite the generally positive responses, several users also identified interface issues or limitations. Some reported unpredictable behaviour of the slider button that is used for confirmation on creating new strategies (jobs):

“That slide when you open a job, it sometimes gets stuck.” (P1)

“[Everything was smooth] except a button that creates a job” (P3)

In addition, a number of participants expressed a desire for enhanced functionality, such as a more detailed portfolio overview, real-time market charts or a wider set of coins in the app.

RQ5: How accessible and user-friendly is the app for individuals with little to no previous experience in finance or behavioural tools?

Participant responses regarding beginner accessibility were mixed. On one hand, some described the app as simple and easy to learn, even without extensive background knowledge. They emphasised that the layout and workflows resembled those of other familiar apps, making the learning curve manageable:

“Actually, you can just learn how to use it in five minutes with not a whole lot of background.” (P1)

“I wasn’t familiar with the app, but after studying each section more carefully, everything became clear.” (P3)

However, other participants expressed some difficulty in understanding specific features, terminology, or advanced settings. This was especially the case when interacting with strategy creation components (jobs):

“Some hints on some financial terms or some crypto terms could be used.” (P2)

Although the application includes static FAQ sections covering key components such as job creation parameters and risk alerting mechanisms, participants experienced confusion during interaction:

“I was wondering why it didn’t appear when I clicked the parameter [in the job creation section].” (P1)

All participants suggested that a more structured onboarding process or educational support features (such as learning sections) would make the app more approachable for beginners:

“It could use some educational features, maybe even some mini courses.” (P2)

“I think it’s a good thing to actually have some learning on those maybe even in the app” (P1)

(Interviewer: Previous people before you have noticed that they would appreciate some learning tools, like learning section. Would you say the app needs something like that?)

“It would be useful”(P3)

Summary of Findings

Thematic analysis of the interviews provided insight into how participants engaged with the application across five core areas.

In terms of behavioural bias detection, users largely perceived the system as accurate (RQ1), with alerts reflecting real-time decision-making patterns. Most participants found the alerts both helpful and behaviourally impactful (RQ2), often prompting reflection, strategy adjustment, or further research before acting, which is the key objective of the application.

Regarding security (RQ3), participants expressed trust in the app’s data protection and authentication mechanisms. Standard features like OTP and biometric login were perceived as sufficient, and no direct concerns were raised.

In terms of design and usability (RQ4), participants generally found the app intuitive and user-friendly, with only minor interface inconsistencies. However, they also identified several areas where key functionality was missing or limited, such as a more detailed portfolio overview, real-time price charts, or a broader range of supported coins, which could declare priorities for future development.

Finally, while the app was generally intuitive for participants with prior financial or crypto experience (RQ5), participants without such a background reported difficulties understanding key features and terminology. In its current state, the application cannot be considered fully beginner-friendly. To support novice users more effectively, a more structured learning approach is needed: onboarding, interactive tutorials, contextual explanations, and dedicated learning sections.

Taken together, the evaluation indicates that the application delivers on its core functionality, especially in behavioural analysis, while also identifying areas for refinement in future development.

Table 4.1*Summary of Findings by Research Question*

RQ	Research Question	Key Findings
RQ1	Accuracy of bias detection	Alerts match user behaviour and were seen as contextually accurate.
RQ2	Usefulness and influence of bias alerts	Alerts help users reflect, adjust behaviour, and conduct further research.
RQ3	Security and trust	Users trust the app due to familiar features and clear data policies.
RQ4	Usability and design	Interface is intuitive, but users noted missing features for deeper engagement.
RQ5	Beginner accessibility	App is difficult for novices.

Conclusion

This thesis studied how behavioural economic biases affect decision-making in cryptocurrency trading and developed a mobile application capable of detecting and mitigating such biases in real time. The project combined an extensive review of behavioural finance theory with the technical design, implementation, and evaluation of a working prototype.

Summary of Research Questions

RQ1: Accuracy of Bias Detection

User feedback shows that the app's bias detection largely aligns with actual trading behaviour. Participants recognised patterns such as FOMO, overconfidence, and loss-related behaviours as accurately reflected in alerts and reports. Some scepticism remained regarding the app's ability to truly infer internal motivation, but overall, the detection system was viewed as contextually precise.

RQ2: Usefulness and Influence of Alerts

Thematic analysis revealed that bias alerts prompted users to reflect on their actions, adjust strategies, and conduct additional research, directly supporting the app's intended purpose. While some users found interventions restrictive during impulsive moments, most appreciated the risk prevention measures.

RQ3: Security and Trust

Participants expressed high levels of trust in the app's data protection, citing features like OTP, 2FA, and biometrics as industry-standard measures.

RQ4: Usability and Design

The app's interface was generally perceived as intuitive, clean and easy to navigate. However, users identified several missing features (e.g., detailed portfolio overview, real-time market charts, a broader coin selection) and some minor interface inconsistencies that limited deeper engagement.

RQ5: Accessibility for Beginners

The application posed challenges for beginners, particularly regarding terminology, functionality explanations, and the strategy creation process. Participants recommended making the app accessible to a wider audience.

Key Achievements

The application's core contribution is the successful integration of behavioural finance theory (overconfidence bias, FOMO, and loss-related biases) into a practical detection and inter-

vention tool. The risk functionality of the app fulfilled all specified requirements (see Table 2.2), reliably identifying key biases, as confirmed by the results of thematic analysis. The application also met the broader requirements set (see Table 3.1), excluding "Secure API Communication", which is addressed in the following section.

Limitations and Unmet Goals

The study's participant group was relatively small, skewed toward young adults (20–30 years old), which could have introduced both sampling and participation biases. All testing was conducted using demo accounts, with no real financial risk involved. This lack of real stakes could have significantly influenced participant risk perception and behaviour, potentially limiting the ecological validity of the findings. Furthermore, because the analysis and interpretation of results were carried out solely by the author, interpretation bias is possible, potentially shaping the conclusions through subjective judgment.

These limit the generalizability of findings, but they also highlight areas for future research and development. More comprehensive studies with more diverse age groups, backgrounds, and experience levels are necessary to validate and extend the application's effectiveness. Multi-analyst approaches to data interpretation would further strengthen future studies. Additional research might also examine the impact of long-term use of the application and the effectiveness of specific mitigation strategies or personalised interventions based on user profiles.

The bias detection capabilities, though functional, are currently limited to a relatively narrow set (confidence, FOMO, and loss behaviour-related). The application would benefit from integrating detection for a wider range of biases.

From a technical perspective, the application's reliance on a trusted network model for internal service communication introduces some security concerns. While this is not an immediate or critical risk in the current context, it is an area that warrants attention in future development. The internal services would benefit from adopting a more robust authentication framework.

Usability testing highlighted that beginner-friendliness remains insufficient, as the application posed challenges for users without prior experience.

Future Areas of Development

Future iterations should focus on making the application more accessible and intuitive for beginner traders by implementing structured onboarding, contextual hints, and learning sections to make the app accessible to a wider audience.

The application would also benefit from expanding the feature set: detailed portfolio overview,

real-time market charts and broader coin selection.

In addition, extending the bias detection module to cover additional patterns, including but not limited to the disposition effect, sunk cost fallacy, gambler's fallacy, mental accounting, or confirmation bias, would provide users with more comprehensive behavioural guidance.

Given recent advances in artificial intelligence, there is a potential to implement AI-powered algorithms, such as machine learning models for trading patterns analysis, anomaly detection techniques to flag unusual activity, or reinforcement learning to adapt feedback to individual users over time, provided user privacy and ethical data collection are prioritised. Specifically, justifications from users with an extreme risk score can be analyzed via Large Language Models (LLMs) to provide in-depth insight into the emotional state of the trader at the time of decision-making. When users initiate new trades despite existing high-risk alerts, their submitted explanations can be analyzed by the AI to uncover recurring patterns of biased reasoning. By considering these insights, users are prompted to identify and correct cognitive biases in their trading decisions.

Subsequently, this emotional analysis can be presented to users as personalized feedback, highlighting signs of potential bias in their own reasoning.

For security improvement, the application needs to adopt a zero-trust network model by implementing short-lived JWTs or digital certificates for internal authorisation of services.

In summary, this work demonstrates the practical value of integrating behavioural economics into digital tools for cryptocurrency trading, while also outlining clear directions for future enhancement and research.

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Appendices

A. Source Codes

The source code for each major application component developed and evaluated in this thesis is publicly available in separate GitHub repositories.

Note on Trade Engine Source Code

The **Trade Engine** component was developed by my supervisor and is not publicly available due to a non-disclosure agreement (NDA). As such, the source code for this module cannot be shared.

Repository Overview

- **Mobile Application (React Native)**
<https://github.com/nikinayzer/tradeguard-mobile>
- **Backend for Frontend (Spring Boot)**
<https://github.com/nikinayzer/tradeguard-server>
- **Risk Detection Module (Python)**
<https://github.com/nikinayzer/tradeguard-health>

Each repository corresponds to the architecture described in Chapters 1 and 3.

B. Demo Video Recording

The video recording of app's functionality and features can be found in the following GitHub repository:

TradeGuard Info

<https://github.com/Nikinayzer/tradeguard-info>

C. Task-based Instructions

TASK-BASED EVALUATION INSTRUCTIONS (TRADEGUARD)

Study Aim

The purpose of this evaluation is to assess whether the application (TradeGuard) functions as intended and provides the essential features required for basic user interaction, risk evaluation, and trading-related operations. This preliminary user study will help identify any usability issues, inconsistencies, or functional failures across key workflows in the application.

Instructions

You are asked to complete a series of tasks designed to simulate common user interactions with the app. These tasks involve creating an account, modifying security settings, interacting with market data, and testing key features such as strategy creation and risk module configuration.

Please complete the tasks in the given order. If you encounter any issues, confusion, or unexpected behavior, you are encouraged to:

- Note it down immediately (use separate freeform document).
- Reach out to the study supervisor at any time.

After completing all the tasks, you are free to use the application during the period of 1-2 days to explore it better.

If you have any questions or notes after completing all the tasks, you will be able to ask/mention these during the interview.

After completing this part (task-based + optional free-roam), please, kindly let the Study supervisor know.

Estimated Time: 60–180 minutes.

Study Supervision and Contact

If you experience technical issues, have questions, or require clarification at any point during the evaluation, please contact:

Nikita Korotov

Email: *korn03@vse.cz*

Task 1: Create an account

1. Open the app and press the **“Sign Up”** button.
2. Enter the required information:
 - Preferred username
 - Valid email address (you may use a temporary email if preferred)
 - Your first and last name
 - Your date of birth
 - Password
3. After registration, attempt to log in using your credentials.
4. You will be prompted to enter a **One-Time Password (OTP)** sent to your email. Enter it to complete the setup.
5. Upon success, you should land on the **Home** screen.

Task 2: Explore the security options of the app

1. Navigate to the **“Profile”** section (via bottom navigation).
2. Open the **“Security”** tab in “Settings” screen. You should find the following options:
 - a) **Change Password**
 1. Attempt to change your password.
 2. After submitting, an OTP will be required to confirm the change.
 3. Log out, then test login with both the **old** and **new** password.
 4. Record the outcomes (e.g., if old password still works, note it).
 - b) **Enable Two-Factor Authentication (2FA)**
 1. Try to enable 2FA.
 2. After enabling, log out and log back in. The app should request an OTP for login.
 3. If it does not, note this behavior.
 - c) **Enable Biometric Authentication ***
 1. If supported on your device (Face ID / fingerprint), enable biometric authentication. *
 2. Minimize and reopen the app. You should be prompted for biometric verification.
 3. Try logging in using biometrics.
 4. Then, fail biometric authentication **10 times** — the app should log you out.
 5. Note any inconsistencies.

Task 3: Set Initial Risk Limits

1. In the “**Profile**” section, open “**Account Limits**” in settings.
2. Review the initial parameters set for you.
3. Are all options and descriptions clear? If not, document what’s unclear.
4. Try to modify any of the parameters manually.

Task 4: Explore the "Market" Section

1. Open the “**Market**” tab via bottom navigation.
2. View the list of available coins.
3. Tap on any coin to see its details.
 - Is the market information accurate? (Compare with external sources: CoinMarketCap, Binance, CoinGecko, etc.)
 - If not, note discrepancies.
4. Check the “**News**” section on the coin detail screen.
 - Are news articles present?
 - Are they recent?
5. Back in the main market list:
 - Use the **star icon** to mark at least **3 coins** as favorites.
 - Confirm these appear in the “**Favorites**” section and on the **Home** screen.

Task 5: Connect your existing exchange account to the app

1. In the “**Profile**” section, locate “**Exchange Accounts.**”
2. Press “**Add Exchange.**”
3. You’ll be asked to provide:
 - API keys
 - Local name for the account
 - Account type (live/demo)
 - Provider (e.g., Binance, Bybit)
4. Attempt to connect an account.
 - Does the app accept the information?
 - Does the account now appear under “Exchange Accounts” in your profile?
 - If not, note the issue.

Task 6: Create an Automated Strategy

1. Open the **“Auto”** tab in the bottom navigation.
2. You should see a strategy builder (referred to as a “job” creator).
 - Refer to the provided **“Strategy Creation Manual”** for parameter definitions.
 - Are all parameters presented? If not, note it down.
3. Configure parameters for **DCA** and **LIQ** modes.
 - Can you adjust job parameters? Are inputs processed correctly?
4. Press **“Browse Available Coins”** and add coins to your strategy.
 - Are coins selectable? Is your selection applied?
 - If not, note issues.
5. Try sliding the **“Create”** button:
 - First, attempt to create a strategy **without selecting any coins** — the app should **prevent this**.
 - Then, submit a complete job.
 - You should see a confirmation message and the job should appear below the form, in the section “Active Jobs”. If not, note it down.
6. Tap on the job created:
 - Check that all job details are displayed correctly.
7. Wait 5–15 minutes:
 - The job should start executing and trigger **push notifications**, which your phone should receive.
 - Check for updates in the **“Portfolio”** tab — you should see positions created by the job.
 - Confirm that job activity and details are correctly shown.
8. Try to create another job and repeat steps 6-7.

Task 7: See your personal risk report

1. Open the **“Health”** tab in the bottom navigation.
2. You should see a screen designed to show your personal risk report.
 - Does it have any “patterns” (referred to as signals or signs in the application)
 - If not, repeat task 6 or lower your personal limits, set up in task 3.
3. Study the report you provided by the application. Do you agree with them? How accurate are they upon your personal judgment? Do you find them accurate? If not, note it down.
4. Try to open more jobs. The risk report should be updated after creating new jobs. Has it increased? Do you find changes in the risk report accurate? If not, note it down.

OPTIONAL Task 1: Connect your Discord account and try to login with it

1. If you have a Discord account that you use, try to connect it to the app.
2. On **Profile screen**, find a button named "Connect Discord Account". Press it.
 - You will be redirected to Discord service, prompting you to login inside Discord.
 - Upon successful login, the app should connect your Discord account to your TradeGuard account.
 - Log out from the app and try to login in app with your Discord account by pressing on "Continue with Discord" button. Repeat the authorization process, after which you should be successfully logged in inside TradeGuard.

D. Interview Questions

This appendix contains the complete set of questions used for the post-use interviews with study participants.

Opening Questions

- Have you used any interesting apps or tools recently?
- If you had to describe your approach to risk in life, not just finance, would you say you're cautious, bold, or somewhere in between?

1. Behavioral Bias Detection & Impact

- Did the app detect any behavioural biases for you during your use?
- Which specific biases were flagged?
- Do you remember what decision or action triggered them?
- Did those detections feel accurate?
- How did the app explain the bias? Was that explanation clear or useful?
- After being warned about a bias, did it influence your next action or decision?
- Would you say the app helped you make more rational or informed decisions?
- Did you ever disagree with the app's assessment of a bias? If so, why?
- Do you think this kind of tool could help you make better decisions?
- If the app failed to detect a bias or made an incorrect assumption, what stood out to you?

2. Security, Privacy & Trust

- Did you feel your personal or financial data was secure while using the app?
- Which security features (e.g., 2FA, OTP, biometrics) stood out to you the most?
- Did the app clearly communicate how your data was being used?
- Were you ever unsure whether your information was stored or shared securely?
- Is there anything that would increase your trust in the app's data handling or security?

3. Design & User Experience (UX/UI)

- What did you think of the app's overall design and layout?
- Were there any annoying elements, clunky interactions, or unclear buttons?
- If you could change one thing about the user interface, what would it be?

4. Usability & Learning Curve

- How easy was it to learn how the app works?
- Were the instructions and onboarding steps sufficient?
- Did anything interrupt your flow or feel hard to figure out?
- Did you feel you needed prior knowledge to use the app effectively?

5. Real-World Applicability & Future Use

- Can you imagine using this app regularly in your financial life?
- Would you be willing to use this app instead of, or alongside, other tools you currently use?
- What would stop you from using the app in real scenarios?
- Would you recommend it to someone? Who, and why?

6. Overall Feedback & Suggestions

- What was the most helpful part of the app for you?
- Did the app meet your expectations? Why or why not?
- If you could suggest one feature or improvement, what would it be?
- Any other thoughts or final feedback you'd like to share?

E. Transcript of Semi-Structured Interview: Participant 1

Transcript of Semi-Structured Interview: Participant 1

<Beginning of the interview>

Interviewer: So, hello [PARTICIPANT'S NAME]. I want to make sure that you have completed the task part of the testing and have completed the consent and background information forms.

This interview will be recorded. All information about your persona will be anonymized. Do you agree?

Participant: Yes.

Interviewer: If you require more time to answer the questions, take all the time you need to prepare.

Interviewer: Let's start with opening questions. Have you used any interesting apps or tools recently for anything, not just finance? If those tools are about finance, I'll be glad if you mention them.

Participant: I traded stocks on Trading 212. I mean, I started, then I withdrew some money, which you probably shouldn't do, but I did. But overall, the tool, the app was quite comfortable, though. A lot of limitations connected with trading financial instruments, like stock. You cannot trade them at any time you want. You need to wait some time so you can withdraw money and everything like that.

Interviewer: If you had to describe your approach to risk in life, not just finance, would you say you're cautious, bold, or somewhere in between?

Participant: I'm everything at the same time.

Interviewer: What do you mean by that?

Participant: There are a lot of parts to life, you can watch that from a multithreaded perspective. For example, your job, the thing you do in your free time, stuff you do with money, and stuff you do with your hands or whatever you create.

You can take huge risks in some parts of your life, take small risks in others, and you can just switch. That's what I do, because I work really quickly.

Interviewer: Let's talk about finance. Do you find yourself more cautious, or sometimes you find yourself actually want to take some risks to achieve fast gains,

past profits?

Participant: As I said, yes. To both of those questions. Ideally, I usually have some bigger portfolio with a smaller amount of money for big risk investments, bigger amount for small risks, and I try to balance them out. Sometimes I like to risk more if that wouldn't hurt me in my financial health. I don't feel like investing long term or with low risk right now is for me. I like to take bigger risks overall.

Interviewer: So, crypto is a perfect place for you, I see.

Participant: Yeah, yeah.

Interviewer: Okay, let's talk about the app itself. I know you used the app for a day, right?

Participant: Yes.

Interviewer: Did the app actually detect any behavioral biases for you during your use?

Participant: Well, considering I was using the demo account, which had a lot of money on it, yeah, I tried to use it. I tried to find out, but eventually it quite swiftly found out that I'm not the safest trader out there. It found just about every risk category it had. I don't know, there were three different ones. I had everything. I think that I should have a medal or something for that. But yeah, it tried to protect me, but what it did, it just annoyed me, so I clicked buttons faster, hoping it would go through the protection as everyone does when there is some time limit. *[note: Participant is talking about cooldown warnings on job creation process, when the risk reported is high]*

Interviewer: You mentioned that you were faced with three categories, like the application detected three categories for you, right?

Participant: Three out of three, I believe.

Interviewer: Which specific biases were flagged?

Participant: Firstly, it detected the FOMO, because I traded one specific coin, was it *ENA* or something, and it linked to its popularity in the news, and it told me that it's a popular commodity right now and you shouldn't trade that. It was like, I don't know, what is it, I googled it and it showed that it's actually predicted to grow or it just was in the news. That was the first one, then I traded some more, tried to do it quickly, and in several

trades it just started saying that I was trading too fast.

The last one actually surprised me, because it was observing how fast do I close my deals. So when I was trying to do day trading and all the stuff like this, it actually noticed and notified that it's alright, but you probably should do something about that. That was actually the one thing that cost me like a thousand dollars. First time I was trading with cryptocurrencies, I think it was many years ago on Binance, and I just was opening the deal the same day I was closing them, I was making some money at first, and then I lost everything. Probably that could have saved me.

Interviewer: Okay. What do you think about those detections? Did they feel accurate?

Participant: It's a thing about them, they're more accurate than what you will ever be able to know about the coins you're trading. For example, about the first one, I did know that was a popular coin, I just saw it, and I, okay, I will buy it, because I want to diversify my portfolio, and it's popular, so why not? And it told me that, yeah, it's in the news and all this stuff, and then I found out about it. So probably they are much more accurate than quite anything you can do as a human.

Interviewer: For example, you mentioned that you were warned about overconfidence, right? You mentioned that you were trading a lot, and the application noticed that you shouldn't trade this much, right? Did this feel accurate?

Participant: I mean, I'm still figuring out how to trade with cryptocurrencies the right way, because when I'm trading stocks, I just buy it and leave it for some few months. With cryptocurrencies, I thought that it was normal to do some weird stuff and buy here, sell here in the course of like a few minutes, but apparently it's not. So, I guess I even learned something from that. Though, yeah, it was quite annoying when it showed me that you were doing something wrong, and I was like, I don't think so, it's not real money [*note: participant tested the app with demo account*].

Interviewer: I see, that's probably the con of testing on demo accounts, right? You should probably have used the real account.

Participant: Well, I wouldn't do that right now, maybe in the future.

Interviewer: Yeah, I see, I see. After being warned about bias, you mentioned you discovered three biases. Did it somehow influence your next action or decision?

Participant: I mean, it's a thing about trading in this app, because it has information that you don't have. I wasn't being able to look at the graphs, because it doesn't have those right now, and to see that something weird is happening with this

cryptocurrency.

And then it warned me about that, and then I was actually able to go to the trading website and see that, yeah, it's like wallets out, 20% a day and all this stuff.

So, yeah, after I found out about it, it was useful, and I think it's a great thing, but probably not for a person like me, because I don't care. I need to base my swift investment decision on something, and you can look at this app as one of the guidelines that you would use, but I don't know where to put it in my head. Like near the projections, news and influencers saying what is going to happen, I still have to make place for this app that is under my hands at the time I'm trading. And it feels a bit weird, I'd say. It can be useful, but I have to get used to it.

Interviewer: Would you say the app helps you make more rational or informed decisions?

Participant: I mean, yeah, it puts a lot of warnings, a lot of countdowns and everything. For example, you need to do some risky trade, and you have to sit here, wait for half a minute, wondering about what the hell you are doing *[note: participant is talking about cooldowns on job creation, when the risk reported is high]*. It gives you some actual space when, for example, you have this situation when you see the coin and then use it and you want to trade it right now, and now you have to actually think about it.

Actually, mentally it makes a bit of a place for you to just wonder. I think it could lead to more informed decisions, but I think more in the long run, when you get used to it and are not annoyed by the stuff it does. I think it truly can.

Interviewer: Have you ever disagreed with the app assessment on bias? And if so, why?

Participant: I mean, this app could not ever know why am I making a decision I am making. If it tells me I am affected by a bias, it couldn't know. It just bases its metrics on some facts, and I don't have facts in my head or some different facts.

Interviewer: That's why it's trying to predict the biases. That's why it can be inaccurate.

Participant: It doesn't predict my biases; it predicts overall financial stuff. It couldn't know why I made those decisions. Overall, I think the application assessments of all this psychological stuff are much more accurate than human, because it actually bases its decisions on some actual facts, as I said.

Interviewer: Let's move to the security part of the application. When using the app, did you feel your personal or financial data was secure while using the app?

Participant: Yep, it feels like a standard app that I don't have a problem trusting to.

Interviewer: Which security features stood out to you the most and what have you tried? For example, have you tried two-factor authentication or have you checked out One-Time Passwords when you tried to change your password or login in the app? Or, for example, if you have used biometric authentication?

Participant: Yeah, I did try all of those. I can't say anything special about those. I think one thing is that this app doesn't force you to do two-factor authentication, which is probably bad for security, but I like it much more than anything else.

Interviewer: So, you would rather be forced to use two-factor authentication?

Participant: I would rather not. I like the switch so I can switch it off and continue with risking my finances even more.

Interviewer: I see.

Participant: I mean, yes, different apps use different stuff. I can imagine there are apps that use some physical things like wallets.

Interviewer: Like USB keys? That's probably too intense, right? That's really annoying sometimes.

Participant: Yeah, but that's probably the thing I like that you can just turn it off.

Interviewer: Did the app clearly communicate how your data was being used?

Participant: Yeah, I think it does a quite standard job about informing how my data was used. No one ever reads those, but I was curious I did. It was quite standard stuff.

I think I trust this app even more because it's not owned by some big company and there's much less of a risk of data leaking out or anything. Even if it's secure, I just trust some small-scale operation much more.

Interviewer: Were you ever unsure whether your information was stored or shared securely?

Participant: I mean, no, because the only secret information that I was using in the app were the keys *[note: API keys]* from the Bybit and they are stored with some over-the-top double security or whatever. It sounds quite secure.

Interviewer: You probably mean encryption, right?

Participant: Yeah, and you can always delete them at any time. I hope nobody will log them.

Interviewer: Well, as a developer I should say that they are definitely not logged.

Participant: It's nice to hear.

Interviewer: Let's move to the design part and the user experience. What did you think of the app's overall design layout?

Participant: It was nice. It was a bit laggy because every app is. I found it actually a bit more comfortable than Binance's app because that one is horrible. I mean, at the time I was using it, I don't know how it is right now. Yeah, it was good. Nothing special.

Interviewer: You mentioned that you're an active user of Invest 212, right?

Participant: Yeah.

Interviewer: If you had to compare them, what stood out to you the most? Are there similarities or something that this app - TradeGuard - is missing?

Participant: You know, TradeGuard has a much smaller scope than Trading 212. I find Trading 212 confusing because there is a lot of stuff happening, a lot of logic inserted somewhere and you don't really know what is happening. I couldn't say something about this, about how the trades are structured in TradeGuard because I'm not really familiar with those types of jobs, DCA and LIQ. I only know the base theory and I was a bit lost. But at the time I was a bit lost, I opened like 10 of them and I think I wasn't supposed to do that. So, I think it's okay.

Interviewer: Well, have you noticed the section that explains the parameters?

*[note: parameters of the job *]*

Participant: Yeah, when I was wondering why it didn't appear when I clicked the parameter. It *[FAQ section]* was right in the upper corner. I mean, you cannot explain those things in the pop-up menu. I should probably buy a book about it and read it. As far as the parameter goes, everything was clear. I didn't find a problem with that.

Interviewer: Okay. Concerning all things said, were there any annoying elements, clunky interactions or unclear items in the application?

Participant: Yeah, the slide that goes halfway and then can stop sometimes and it

doesn't go back every time and it's a funny thing.

Interviewer: What do you mean by that?

Participant: Well, that slide when you open a job, it sometimes gets stuck.

Interviewer: We should probably have fixed that.

Participant: Probably yes, but it's just something funny in a trading-oriented app, you don't always have those.

Interviewer: Would you rather want to keep this slider, but fixed, or a standard confirmation button?

Participant: Yeah, you can keep it. The simple button is even more boring. You can't be more boring than that. Sliders are something. I like slider more.

Interviewer: Okay, I see. That's something you want to change in the app. Is there anything else you would like to change in the user interface?

Participant: The graphs, of course.

Interviewer: Can you elaborate?

Participant: The graphs of current currencies because it lacks those right now.

Interviewer: You mean the coin overview section, right?

Participant: Yes.

Interviewer: When you try to open the coin, there are no graphs, right? There is just information.

Participant: Maybe I would want to have some more extensive view of my portfolio. Like in Trading 212, I have this tile layout where I can see visually how big my investment in that stock is and how big and different one.

I would like that one because when I open many jobs and many coins, firstly, I have no idea where the progress is. It would be a nice thing to actually do a projection of what your portfolio will be and do that tile projection right now. It lacks those in a quite useful way.

Interviewer: That's actually a nice recommendation. Thank you.
Let's move to the next part. The questions will be about usability and learning curve.

How easy was it to learn how the app works? I know you're partly answering this question, but... *[Interviewer was interrupted]*

Participant: I mean I was supposed to use "Automate" section *[note: job creation part of the app]* right about the time I was wondering where do I do it manually? Where can I invest a million dollars in Trump coin or anything? But yeah, there's not much to it and actually you can just learn how to use it in five minutes with not a whole lot of background.

Interviewer: Were you using any instructions or have you noticed and considered onboarding steps? Were they sufficient? Like there are a few onboarding tips on the application, have you noticed them?

Participant: The application, you know the thing about the good design is you don't notice good design. You only notice the bad design and here you just go register, you go fill out your info and it works. Which is nothing special, but it is special because it's good design, you don't notice it and you don't even have the need to go to use some instructions to how to do something. It just works and it's really good.

Interviewer: What do you think, is some prior knowledge needed to use the app effectively? Like, you have financial background, if I'm right. So that's probably helped you a lot in the application?

Participant: I mean, it doesn't help much in the application I'd say, because the financial world is a big world and even though I work in a bank, on a day to day basis we interact with different things that probably some other people interact with, and here in the app we use some different instruments. So, I think it's a good thing to actually have some learning on those maybe even in the app. So maybe some learning section, I don't know.

But I think it's more required for actual investments because if you don't know anything about investments you just lose all your money even though you have all those warnings and everything that wouldn't save you if you don't know anything.

Interviewer: Agree. So, you suggest that the app probably in the future should have something like a learning section?

Participant: I actually use some little more complex financial instruments. Here, you buy cryptocurrencies, some amount, not just all at once, right now. You can do that on Binance or Bybit, here it's a bit more complicated and a bit more disconnected from the user. I think firstly you have to learn what those instruments are and secondly how to use them effectively. I think that's an important thing.

Interviewer: Ok, I see, that's helpful. Can you imagine using this app regularly in

your financial life? Like, would you consider this app when you actually want to invest in crypto?

Participant: I think I actually wouldn't be investing any other way because I had my share of losing and everything, and this app actually feels like it would give you some protection and some protection is much, much better than none. Like in Binance, when you have like 200x shoulders.

Interviewer: Binance doesn't even have this feature, right? *[note: Interviewer means risk reports]*

Participant: Yeah, it doesn't even have this feature and you're safe here. Cryptocurrency is a wild world, and was before, still is, and to have literally anything to tell you except other people that you can't really trust to let you know what you're doing, how bad is it or how good is it. It's a great thing.

Interviewer: So, would you use this app standalone if you want to invest in crypto or would you use it alongside other tools? For example, using Bybit or Binance for market information or other tools, if you have any in mind?

Participant: I mean, I don't know. It depends on your level of expertise which I would say don't have that big of a level, I'm no trader. I don't draw arrows on the graphs and trying to find out what's going to happen next, and I don't read a lot of news. But just to have all of those tools *[note: DCA and LIQ jobs]* with some safeguards at the same place will be enough for 80% of people. If you just use this thing without anything else, like the premise that it will be completed, I think it can be used standalone for sure.

Interviewer: What would stop you from using the app in real scenarios?

Participant: What do you mean?

Interviewer: For example, let's imagine that you're doing daily trading with this application. Can you imagine an event that would just interrupt you and you would reconsider using this app anymore? For example, data leaks or server breakdown?

Participant: I think if there's something happening on the market I would like, for example, to close my position fast. I don't think it allows for this right now and I don't think it's the use case of the app but so be it. I really wouldn't like to be in a situation where the servers are down. It wouldn't really be too pretty. But I don't know, I still have to adjust to this. For example, when I see the market going down, I want to sell. I don't do this because I tell myself that it's alright but sometimes I just want to do it, and I don't know if it allows for this.

Participant: So, you would appreciate an option to ...

[Interviewer was interrupted. Note: liquidate portfolio positions quickly]

Participant: It kind of defeats the whole purpose, if you allow an option to just do trades like this. I don't know. As I say I'm no expert. I'm just thinking out loud. But I think in the current shape it's a really interesting instrument that you of course have to try for long term to have any real conclusions.

Interviewer: Will you recommend it to someone?

Participant: Well, I mean for me. I would actually like to have some kind of app like this for trading stocks. I don't think Trading 212 allows this. Buying stocks over the course of some period. I think it does, but it's just not the same. I would recommend it for me and probably for everyone that I know who's trading crypto which is precisely zero people. So, I guess just for me.

Interviewer: I see. What was the most helpful part of the application for you? What did you like the most? For example, the market section when you were able to gather market information, or the auto trading section when you were able to open jobs or health screen, where the app reports your risks?

Participant: You know, when you're bored you go sometimes complete some psychological tests for fun with friends. You get through 100 questions and in the end, you have assessment of you. That's what you're there for. I like the risk section about that, because you actually can go to some place and see somewhat accurate assessment of your trades and what you're doing, and it kind of changes your perspective a bit and is really helpful I think. And I liked it a lot because it told me something about myself I didn't know. It told me that I'm a [PROFANITY] in the trading.

Interviewer: I am definitely going to censor this one. Did the app meet your expectations? If you had any.

Participant: I had an expectation for it to be something new and I think it fulfilled those. As I said in all of my previous answers.

Interviewer: Any other thoughts or final feedback you would like to share?

Participant: Completed, I would like to test it in a real-world scenario *[note: with real money]*.

I think it's a good tool and I would like to find out how it works in the long term. I want someone else to use it and write some things on the internet that I can read and look

at it. That's the thing I want right now. I would like to see it released and used.

Interviewer: Thank you for your interview. Thank you for your time. That's the end of the interview. It was nice to talk with you.

Participant: Thank you very much.

<End of the interview>

F. Transcript of Semi-Structured Interview: Participant 2

Transcript of Semi-Structured Interview: Participant 2

<Beginning of the interview>

Interviewer: Hello, [PARTICIPANT'S NAME].

Participant: Hello.

Interviewer: I want to make sure that you have completed task-based part of the testing and have completed the consent and the background information form.

Participant: Yes.

Interviewer: This interview will be recorded. All information about your persona will be anonymized. Do you agree to continue?

Participant: Yes.

Interviewer: I have to notice that if you need more time to answer the questions, take all the time you need to prepare.

Participant: Yes.

Interviewer: I will start with generic questions: have you used any interesting tools recently? For anything, not only finance.

Participant: Well, I was buying bonds via interactive broker application. I was also using Binance application for buying stablecoins like USDT. I also bought Bitcoin through Binance application. I was also testing Bybit application recently.

Interviewer: Oh, so you just found out about Bybit? Why did you switch from Binance to Bybit?

Participant: Because of regulations of Binance, not all coins or trading operations are available there. Like, you can't use Russian Rubles there anymore. So, I had to use other instruments as alternatives, for example, Bybit.

Interviewer: I see, it makes sense. If you had to describe your approach to risk in life, not just finance as well, would you say you're cautious, bold or somewhere in between?

Participant: That depends on how much I invest. If I have a feeling something will grow, I can invest some small amount into some really risky instrument or into some risky coin or currency. But if I am investing for a long term, I use higher amounts and some more conservative instruments and currencies with lower risks.

Interviewer: I see. I think we should start with questions about my application. Did the app detect any behavioral biases for you during your reviews? And before you answer that, I should ask how much exactly have you used the app?

Participant: Well, I tried to start some jobs with some coins I've heard about in the news or on social media, and it detected some biases, because I've used some so-called "meme" coins, though it's not mentioned in the app, those are called so. But people call those coins "meme" coins. And when I started trending them, it detected biased behavior.

Also, I was buying TON coin there, and it was so-called hype, because in the news it was promoted as a promising coin. So, I bought that, and the app detected that behavior and supposed that I was buying it because it was hyped and warned me about it.

So yes, some biases were detected.

Interviewer: Do you agree that detection this detection of TON coin was accurate?

Participant: Yeah, I would say so.

Interviewer: Are there any other biases that app has flagged for you?

Participant: Well, it also detected that I was buying too much and too quick, because I started jobs with a lot of coins and a lot of illiquidity, and the app warned me about that, that it was a very risky operation.

Interviewer: What exactly has it warned you? Just because you started operations with a lot of coins, or were you invested in an illiquid coin?

Participant: I think it detected that I was buying a lot of different coins, and I was spending a lot of money on it.

Interviewer: What exactly did the app say to you?

Participant: I think it detected burst of trading. I was investing so much that I was out of my limits, which I set in the settings of the app in my account, and when I was overstepping this limits the app warned me that I was in a very risky behavior.

Interviewer: So, I assume that the app did explain the biases for you well and the

explanation was clear?

Participant: Yeah, the explanation was clear and based on the settings.

Interviewer: After being warned about a bias, for example, you mentioned TON hype detection or burst of trading, right? Did it influence your next action or decision?

Participant: Yes, I decided to do research on what I'm buying and why the app is warning me about my behavior. So, I decided to find out more about my actions.

Interviewer: I assume you're talking about TON case. But you also mentioned that the app detected burst of trading, right? Did the explanation change your decision or decision-making?

Participant: Yes, I decided to slow down and take my time to think more about if I should do what I was doing.

Interviewer: So I assume that app actually helped you to make more rational or informed decisions, right?

Participant: Yes, exactly. Because before I was warned, I didn't think about that notch that it was a risky behavior, but if the app says so, I decided to take my time and do a research.

Interviewer: Do you think there were, as well, any other biases that app did not detect for you?

Participant: Well, as I said before, there are some coins that are called "meme" coins, and they are existing purely as a joke. They don't serve any utilitarian purpose. And when people are buying those, I think it might be a risky behavior, but that's just my opinion.

Interviewer: I assume some people just want to invest short-term and just obtain huge gains on these "meme" coins. I mean, if you invest in the long term, that's a problem. But if your initial aim is just to make some quick profit, I mean, why not?

Participant: Yes, but that's risky behavior. And I think people should be warned about it. Yes, they should be allowed to do that in the app. App can warn about it, but people still should be allowed to do that, which is the way the app is behaving, and then that's right.

Interviewer: Okay, I see. Let's move to the security topic. Did you feel your personal

or financial data was secure while using the app?

Participant: I think, yes. The app looked like those very popular other apps. And I felt like I can trust this app because it follows some certain patterns other apps follow.

Interviewer: Have you tried all the security features? For example, the app has two-factor authentication, one-time passwords for login or operations like changing your password. Have you used them? And if so, what did stand out for you the most?

Participant: Well, I've tried to factor authentication. I've changed my password one time. I've also connected another account from social network, Discord. So, yes, there are some measures to make your account safer, and I've used some of those options.

Interviewer: Did the app clearly communicate how your data was being used?

Participant: Yes, it was very clear to me because when I was registering an account there was a notification about how my data would be used. It was very clear to me since that moment.

Interviewer: Were you ever unsure whether your information was stored or shared securely?

Participant: No, all the information I provided to the app was stored in an encrypted way, so I wasn't worried about it.

Interviewer: So, you think that because the app said you so, right?

Participant: Yes.

Interviewer: I see. Is there anything else that would increase your trust in the app's data handling or security?

Participant: The app suggested me some improvements in the security of my account, like using biometric authentication or two-factor authentication. After I implemented those, I was sure it was all secure.

Interviewer: What about design? What did you think of the app's overall design and layout?

Participant: I think it was very clear to me and *the app's design* follows the patterns of other apps, so it was very intuitive to use. I think it was good.

Interviewer: Were there any annoying elements, clunky interactions or unclear buttons?

Participant: Well, I think some notifications were excessive and I didn't really need them.

Interviewer: If you could change one thing about the user interface, what would it be?

Participant: There is an option of showing me my portfolio and maybe some basic information about the positions in my portfolio, but it was, as I said, very basic. I would like app to provide me more information on my portfolio. Also, I think that since the app helps to prevent risky behavior, it will be beginner oriented. So, some hints on some financial terms or some crypto terms could be used, like you click on the question mark and the app explains you what this or another button does or what this or another operation will do.

Interviewer: I see, that could actually improve the app, I agree. What about usability?

How easy was it to learn how the app works? Have you had any issues with onboarding on the app?

Participant: No, because, as I said, this app in its design and its interface was very similar to other financial apps, so it was very intuitive. I knew where I should go and what I should click to get what I need.

Interviewer: Were the instructions and onboarding steps sufficient for you?

Participant: Yes.

Interviewer: Did anything interrupt your flow or feel hard to figure out? For example, when you had to create jobs, was it intuitive for you?

Participant: Well, I think for me, it wasn't clear about some things the app can do, for example, what do steps mean? When I change the amount of steps, what does it mean? That was unclear. As well as other financial stuff, because they would need to be explained for a beginner. But if I knew about those kinds of financial operations, I would be okay with this, because the app is intuitive.

Interviewer: So, you are basically saying that you need some prior knowledge to use the effectively?

Participant: Yes, exactly.

Interviewer: Let's move to the most interesting part. If you had to think about real world applicability and future use, can you imagine using this up in your financial life?

Participant: Yes, I think I can be interested in crypto, but I don't know much about it. I don't know much about the financial world in general. I could use some beginner-friendly application, which would warn me about some risky behavior, and also explain to me some basic tools. I think this app could have a future in my financial life.

Interviewer: Assuming this app will continue, you would expect this app to implement some more user-friendly features, like some education on crypto, or something like that?

Participant: Yes, exactly. It could use some educational features, maybe even some mini courses.

Interviewer: Mini courses on crypto, or behavioral theory, or what part of education content?

Participant: I think it's all interconnected, so both on risky behavior and crypto, because the user will need to know about both.

Interviewer: What would stop you from using the app in real scenarios, assuming the current state of the application?

Participant: I think maybe some technical issues could prevent that. If the app wouldn't behave in the way I would like it to behave, or if it would be too slow, or if the app wouldn't be secure enough, because for financial apps, it's really important to be secure, and if my financial data would be leaked, I wouldn't be using this app anymore for sure.

Interviewer: Would you be willing to use this app instead of, or alongside other tools you currently use, for example, Binance or Bybit, or other tools you're using?

Participant: Yes, I could start using it instead of Binance, because Binance is not beginner friendly at all. It will need much, much more prior knowledge compared to this app.

Interviewer: Okay, that's great. Let's move to the overall feedback part. What was the most helpful part of the app for you? What did you appreciate the most, the

possibility of opening half-automated jobs/strategies, or risk analysis, or portfolio overview, or anything else?

Participant: Yes, I think the automated jobs is a nice feature, which could save you time. Also, the risk analysis isn't something I've met in other apps, so that was a nice feature.

Interviewer: Did the app meet your expectations? If so, why, or why not?

Participant: Well, my expectations were that this app should warn me on risky behavior and analyze my financial behavior overall, which it did. So yes, it met my expectations from my point of view as a beginner.

Interviewer: Do you have any other thoughts or final feedback you'd like to show?

Participant: Yes, I'd like to see there more coins available in the app. Also, as I mentioned before, I'd like to have more information on my portfolio. And also I've noticed that there is no spot trading in the app. I would like also see that introduced.

Interviewer: I see. Well, thank you for the interview. That was nice to hear. We'll process all the information you have provided to us and improve the app. Thank you for the participation!

Participant: Thank you.

Interviewer: Bye!

<End of the interview>

G. Transcript of Semi-Structured Interview: Participant 3

Transcript of Semi-Structured Interview: Participant 3

<Beginning of the interview>

Interviewer: Hello, [PARTICIPANT NAME]. I want to make sure that you have completed the task part of testing and have completed consent and background information forms. This interview will be recorded. All information about your persona will be anonymized. Do you agree with that?

Participant: Yes.

Interviewer: Have you completed the task-based part and background information forms?

Participant: Yes.

Interviewer: If you require more time to answer the questions, take all the time you need to prepare. Let's start with the opening questions. And I want to ask you if you have used any interesting apps recently for finance? But not specifically finance, if there is anything else you want to mention, make sure to mention it.

Participant: I use Bybit every day for selling crypto, especially USDT in P2P. *[note: P2P (peer-to-peer) trading lets users exchange crypto directly, usually via local payment channels and negotiated prices.]*

Interviewer: P2P, okay, I see. Have you used other financial tools like bank applications, or maybe Binance?

Participant: I used Binance, but it was a year ago. Now I use only Bybit for crypto trading.

Interviewer: And other financial apps, like investing in bonds, like Trading 212 or other investing apps?

Participant: No, only my bank app in Czech.

Interviewer: If you had to describe your approach to risk in life, not just finance, would you say you're cautious, bold, or somewhere in between?

Participant: I think I'm cautious, because I try to think through every decision before making it.

Interviewer: Like, you're analyzing the situation, trying to think of all the possibilities, and after that you make a decision, right?

Participant: I'm trying.

Interviewer: So, you're not the one who's known for just taking unthought decisions, right?

Participant: No.

Interviewer: Okay. For how long you have used the TradeGuard app?

Participant: I used it in my free time for two days, I think.

Interviewer: I see. So, you used it for two days. Did the app detect any behavioral balances for you during your use?

Participant: Yes, they were overconfidence and FOMO.

Interviewer: Do you remember what decision or actions triggered them?

Participant: Initially, when I was introduced to the app and looked at all available coins and possibilities, I wanted to try it all. I created a lot of jobs with various coins and settings, and then I received a notification saying that apparently my risk score is high, and then I opened health report, and it was saying that I violated my limits, and that resolved in high overconfidence risk score and FOMO. In the case of FOMO, the reason was overhype around the XRP coin, and I only found that about thanks to the app.

Interviewer: Did those detections feel accurate for you?

Participant: Yes, I think the detections were accurate. I wasn't consciously aware of those biases at the time, but when I reviewed the feedback, it seemed to match the context of my actions and made sense.

Interviewer: And you mentioned that you were triggered with FOMO, right?

Participant: Yes.

Interviewer: How did the app explain the bias? Was the explanation clear or useful? Or you just noticed that you have higher FOMO risk and just tried to figure it out on your own?

Participant: I think the app showed the name of the bias and advised something reconsidering my trading positions. No, that's all.

Interviewer: Okay, and after being warned about the bias, did it influence your next action or decision?

Participant: I stopped creating jobs after receiving a warning about overconfidence, and after the app notified me about FOMO and the overhype surrounding XRP, I chose to research the topic further on the Internet.

Interviewer: And by saying topic, what exactly do you mean? Researching FOMO, the effect on its own?

Participant: No, researching something about XRP.

Interviewer: Aha, I see. So, it made you rethink your investment in the XRP, right?

Participant: Yes.

Interviewer: Would you say that the app helped you make more rational or informed decisions?

Participant: Hm, yes, sure.

Interviewer: So, you think stopping your trading activity after being warned about overconfidence was your rational decision?

Participant: Yeah.

Interviewer: Okay, I see. Did you ever disagree with the app assessment of a bias, and if so, why?

Participant: No.

Interviewer: And for example, XRP. Just because I'm a developer, I know that sometimes this hype detector can be not lying, but exaggerating things. Like, if coin is marked as being "hyped", it doesn't necessarily mean it's wrong to invest, or it's a not smart move. Sometimes it just can be exaggerated. In your case, do you think this could take a place?

Participant: No.

Interviewer: Do you think this kind of tool could help you make better decisions?

Participant: Not only could, but it already helped, and the app warned me when I started acting irrationally.

Interviewer: Has the app ever failed to detect a bias or made an incorrect assumption? For example, there is something you probably knew about yourself, but the app did not detect it? Or when you showed obvious signs of some bias, for example, making a lot of trades, but the app did not warn you about it?

Participant: No.

Interviewer: I see. Let's move to another part. Those will be questions about security. When using the app, did you feel your personal and financial data was secure while using the app?

Participant: Yeah, because I had no reason to think it might be insecure.

Interviewer: I see. Which security features, for example, the app has two-factor authentication, one-time passwords, or even biometrics for local authentication; Which security features stood out to you the most?

Participant: I think the biometrics feature, because it's not as common in apps as something like OTP, and it gave me an understanding that the app really cares about my privacy and security.

Interviewer: I see. Did the app clearly communicate how your data was being used?

Participant: Yes, it was stated during creation an account that my personal data is used solely for processing within the app and is not shared anywhere else, and the same applied to API keys when connecting an exchange account saying they are encrypted and used only in internal system.

Interviewer: Were you ever unsure whether your information was stored or shared securely?

Participant: No.

Interviewer: What makes you think so?

Participant: I had no reason to think it might be unsecured.

Interviewer: I see. Is there anything that would increase your trust in the app's data handling or security? For example, are there any features you want to see in the application or is it good enough for you already?

Participant: Now, in the moment, I have no idea.

Interviewer: You can take your time to think about it.

Participant: I don't want to.

Interviewer: Okay, let's move to the next part. Now we'll talk about the design and overall app's feeling. What do you think of the app's overall design and layout? Was it clear, useful, intuitive?

Participant: Yeah, I think it looks simple and neat, and everything was in its place, was user-friendly and visually appealing. It was good.

Interviewer: Were there any annoying elements, clunky interactions or like, for example, elements of design or buttons?

Participant: Yeah, everything worked smoothly except a button that creates a job, where you had to swipe and sometimes it got stuck.

Interviewer: You're not the first person actually talking about the button and I want to ask you: if you'd rather prefer a normal button or keep this one but fix it?

Participant: This button is fine. So, I think just fixed.

Interviewer: I see. If you could change one thing about the user interface, what would it be? If there's nothing you want to change, feel free to just skip this question.

Participant: No, only this button.

Interviewer: Let's talk about usability and about learning curve. How easy was it for you to learn how the app works and just get into this onboarding process, get through this onboarding process?

Participant: It wasn't easy at first. I wasn't familiar with the app but after studying each section more carefully, everything became clear. So, it takes time.

Interviewer: And if you had to elaborate on that, was the job creation part hard or health screen hard to understand how it works? What specific part of the app was hard to understand?

Participant: I think it was a job creation part because I don't have much experience in crypto trading, and some options were not clear to me. And even with the help screen, it takes time to understand.

Interviewer: But after all, you understood all the options, right?

Participant: Yeah.

Interviewer: Did anything interrupt your flow or feel hard to figure out besides the job creation part?

Participant: No.

Interviewer: I'll take it. You already kind of answered this question partly, but let me ask you once again: did you feel you need prior knowledge to use the app effectively?

Participant: Yeah, definitely some prior knowledge about trading and biases is required.

Interviewer: Okay, can you imagine using this app regularly in your financial life?

Participant: Yes, if I decide to actively engage in crypto trading then definitely yes.

Interviewer: Would you be willing to use this app instead of or alongside other tools you currently use and I mean financial tools?

Participant: Alongside other tools, because I like to have a lot of tools, and now I have Binance, Bybit, Kucoin.

Interviewer: What would stop you from using this app in real scenarios if you had to engage in crypto trading?

Participant: It could be something like data leaks because I remember Bybit was data leaked *[note: participant means Bybit's hacking incident that happened in the begging of year 2025]* this year and lost a lot of coins.

Interviewer: So, security is important for you?

Participant: Sure.

Interviewer: Would you recommend this app to someone like your friends or family,

siblings?

Participant: Why not, yes. I can recommend it for my dad because he trades crypto.

Interviewer: What was the most helpful part of the application for you?

Participant: I think it was the health section, because I'm cautious and I don't want to make risky decisions, and this part of the app can warn me if I do something risky.

Interviewer: If you could suggest one feature or improvement, what would it be?

Participant: I think more explanations about biases, and ... *[participant hesitates]*
Everything else is fine.

Interviewer: And you mentioned the app probably is not the most beginner friendly, right?

Participant: Yes.

Interviewer: Previous people before you have noticed that they would appreciate some learning tools, like learning section. Would you say the app needs something like that?

Participant: Yeah, it would be useful.

Interviewer: Any other thoughts or final feedback you'd like to share?

Participant: No.

Interviewer: I see. I appreciate it. This is the end of the interview. Thank you, a lot, for your participation.

Participant: And thank you too for this useful app.

Interviewer: I'll let you know when the results are done. Thank you and bye.

<End of the interview>

H. API Documentation of BFF Service

BFF API

Overview

This API specification describes the Backend-for-Frontend (BFF) service, providing streamlined, secure endpoints for mobile application and internal services.

Paths

GET /events/stream

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

GET /users/{userId}

Parameters

Type	Name	Description	Schema
path	userId <i>required</i>		integer (int64)

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

POST /users/{userId}

Parameters

Type	Name	Description	Schema
path	userId <i>required</i>		integer (int64)

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

POST /users/{userId}/exchange-accounts/{id}/update

Parameters

Type	Name	Description	Schema
path	id <i>required</i>		integer (int64)
path	userId <i>required</i>		integer (int64)

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

POST /users/{userId}/exchange-accounts/add

Parameters

Type	Name	Description	Schema
path	userId <i>required</i>		integer (int64)

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

GET /users/{userId}/limits

Parameters

Type	Name	Description	Schema
path	userId <i>required</i>		integer (int64)

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

POST /users/{userId}/limits

Parameters

Type	Name	Description	Schema
path	userId <i>required</i>		integer (int64)

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

POST /users/{userId}/settings/security/update

Parameters

Type	Name	Description	Schema
path	userId <i>required</i>		integer (int64)

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

DELETE /users/{userId}/exchange-accounts/{id}/delete

Parameters

Type	Name	Description	Schema
path	id <i>required</i>		integer (int64)
path	userId <i>required</i>		integer (int64)

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

GET /users

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

GET /users/{userId}/exchange-accounts

Parameters

Type	Name	Description	Schema
path	userId <i>required</i>		integer (int64)

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

GET /users/{userId}/exchange-accounts/{id}

Parameters

Type	Name	Description	Schema
path	id <i>required</i>		integer (int64)
path	userId <i>required</i>		integer (int64)

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

GET /users/{userId}/settings/security

Parameters

Type	Name	Description	Schema
path	userId <i>required</i>		integer (int64)

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

GET /users/{userId}/pushTokens

Parameters

Type	Name	Description	Schema
path	userId <i>required</i>		integer (int64)

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

POST /jobs/dca

Parameters

Type	Name	Description	Schema
header	X-Platform-Type required		string

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

POST /jobs/liq

Parameters

Type	Name	Description	Schema
header	X-Platform-Type required		string

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

POST /jobs/{id}/pause

Parameters

Type	Name	Description	Schema
header	X-Platform-Type <i>required</i>		string
path	id <i>required</i>		integer (int64)

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

POST /jobs/{id}/resume

Parameters

Type	Name	Description	Schema
header	X-Platform-Type <i>required</i>		string
path	id <i>required</i>		integer (int64)

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

POST /jobs/{id}/stop

Parameters

Type	Name	Description	Schema
header	X-Platform-Type <i>required</i>		string
path	id <i>required</i>		integer (int64)

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

POST /jobs/{id}/cancel

Parameters

Type	Name	Description	Schema
header	X-Platform-Type required		string
path	id required		integer (int64)

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

GET /jobs

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

GET /users/jobs

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

GET /users/jobs/active

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

GET /users/jobs/completed

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

GET /jobs/user/{userId}

Parameters

Type	Name	Description	Schema
query	timeframe <i>optional</i>		integer (int32)
path	userId <i>required</i>		integer (int64)

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

GET /jobs/{id}

Parameters

Type	Name	Description	Schema
path	id <i>required</i>		integer (int64)

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

POST /internal/notifications/send/maintenance

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

GET /internal/connections/discord/{discordId}

Parameters

Type	Name	Description	Schema
path	discordId <i>required</i>		integer (int64)

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

GET /internal/users/{userId}/limits

Parameters

Type	Name	Description	Schema
path	userId <i>required</i>		integer (int64)

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

POST /auth/register

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

POST /auth/login

Parameters

Type	Name	Description	Schema
header	X-Push-Token <i>optional</i>		string

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

POST /auth/verify-otp

Parameters

Type	Name	Description	Schema
header	X-Push-Token <i>optional</i>		string

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

POST /auth/password-change

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

POST /auth/password-change/verify

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

POST /auth/logout

Parameters

Type	Name	Description	Schema
header	X-Push-Token <i>optional</i>		string

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

GET /auth/validate

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

POST /auth/discord/exchange

Parameters

Type	Name	Description	Schema
header	X-Push-Token <i>optional</i>		string

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

GET /news

Parameters

Type	Name	Description	Schema
query	page <i>optional</i>		integer (int32)

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

GET /news/{coin}

Parameters

Type	Name	Description	Schema
query	page <i>optional</i>		integer (int32)
path	coin <i>required</i>		string

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

POST

/users/{userId}/notifications/{notificationId}/read

Parameters

Type	Name	Description	Schema
path	notificationId <i>required</i>		integer (int64)
path	userId <i>required</i>		integer (int64)

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

GET /users/{userId}/notifications

Parameters

Type	Name	Description	Schema
path	userId <i>required</i>		integer (int64)

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

GET /users/{userId}/notifications/unread

Parameters

Type	Name	Description	Schema
path	userId <i>required</i>		integer (int64)

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

Code	Description	Links
200	OK <i>Content</i> /	No Links

GET /market

Parameters

Type	Name	Description	Schema
query	counter <i>required</i>		string
query	base <i>required</i>		string

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

GET /market/all

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

GET /market/tokens

Parameters

Type	Name	Description	Schema
query	symbols <i>required</i>		< string > array

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

GET /market/categories

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

GET /market/available-tokens

Responses

Code	Description	Links
200	OK <i>Content</i> /	No Links

Components

Schemas

SseEmitter

Properties

Name	Description	Schema
timeout <i>required</i>		integer (int64)

UserUpdateRequestDTO

Properties

Name	Description	Schema
email <i>optional</i>		string

Name	Description	Schema
firstName <i>optional</i>		string
lastName <i>optional</i>		string

Instant

DiscordAccountDTO

Properties

Name	Description	Schema
discordId <i>optional</i>		string
username <i>optional</i>		string
discriminator <i>optional</i>		string
avatar <i>optional</i>		string

ExchangeAccountDTO

Properties

Name	Description	Schema
id <i>optional</i>		integer (int64)
provider <i>optional</i>		string
name <i>optional</i>		string
readWriteApiKey <i>optional</i>		string
readWriteApiSecret <i>optional</i>		string

UserDTO

Properties

Name	Description	Schema
username <i>optional</i>		string
email <i>optional</i>		string

Name	Description	Schema
firstName <i>optional</i>		string

ExchangeAccountUpdatedDTO

Properties

Name	Description	Schema
name <i>optional</i>		string
readWriteApiKey <i>optional</i>		string
readWriteApiSecret <i>optional</i>		string

ExchangeAccountCreationRequestDTO

Properties

Name	Description	Schema
provider <i>optional</i>		string
name <i>optional</i>		string
demo <i>optional</i>		boolean
readWriteApiKey <i>optional</i>		string
readWriteApiSecret <i>optional</i>		string

UpdateUserAccountLimitsRequestDTO

Properties

Name	Description	Schema
maxSingleJobLimit <i>optional</i>		string
maxDailyTradingLimit <i>optional</i>		string
maxConcurrentOrders <i>optional</i>		string
maxDailyTrades <i>optional</i>		string

Name	Description	Schema
tradingCooldown <i>optional</i>		string
maxPortfolioRisk <i>optional</i>		string
allowDcaForce <i>optional</i>		boolean
allowLiqForce <i>optional</i>		boolean
dailyLossLimit <i>optional</i>		string
maxConsecutiveLosses <i>optional</i>		string
maxDailyBalanceChange <i>optional</i>		string
volatilityLimit <i>optional</i>		string
liquidityThreshold <i>optional</i>		string

BigDecimal

UserAccountLimitsDTO

Properties

Name	Description	Schema
id <i>optional</i>		integer (int64)
userId <i>optional</i>		integer (int64)
maxSingleJobLimit <i>optional</i>		BigDecimal
maxDailyTradingLimit <i>optional</i>		BigDecimal
maxPortfolioRisk <i>optional</i>		BigDecimal
maxConcurrentOrders <i>optional</i>		integer (int32)

Name	Description	Schema
maxDailyTrades <i>optional</i>		integer (int32)
tradingCooldown <i>optional</i>		integer (int32)
dailyLossLimit <i>optional</i>		BigDecimal
maxDailyBalanceChange <i>optional</i>		BigDecimal

UserSecuritySettingsDTO

Properties

Name	Description	Schema
lastPasswordChangeDate <i>optional</i>		string
twoFactorEnabled <i>optional</i>		boolean

PushToken

Properties

Name	Description	Schema
id <i>optional</i>		integer (int64)
userId <i>optional</i>		integer (int64)
token <i>optional</i>		string

DcaJobSubmissionDTO

Properties

Name	Description	Schema
amount <i>optional</i>		number (double)
name <i>optional</i>		string
coins <i>optional</i>		< string > array
side <i>optional</i>		string
discountPct <i>optional</i>		number (double)

Name	Description	Schema
totalSteps <i>optional</i>		integer (int32)
durationMinut es <i>optional</i>		number (double)
source <i>optional</i>		string

LiqJobSubmissionDTO

Properties

Name	Description	Schema
amount <i>optional</i>		number (double)
name <i>optional</i>		string
coins <i>optional</i>		< string > array
side <i>optional</i>		string
discountPct <i>optional</i>		number (double)
totalSteps <i>optional</i>		integer (int32)
durationMinut es <i>optional</i>		number (double)
source <i>optional</i>		string

Job

Properties

Name	Description	Schema
jobId <i>optional</i>		integer (int64)
userId <i>optional</i>		integer (int64)
strategy <i>optional</i>		enum (DCA,LIQ)
status <i>optional</i>		enum (CREATED,PAU SED,IN_PROGR ESS,STOPPED,C ANCELED,FINI SHED)
stepsDone <i>optional</i>		integer (int32)

Name	Description	Schema
coins <i>optional</i>		< string > array
side <i>optional</i>		string
discountPct <i>optional</i>		number (double)
amount <i>optional</i>		number (double)
stepsTotal <i>optional</i>		integer (int32)
durationMinut es <i>optional</i>		number (double)
createdAt <i>optional</i>		Instant
updatedAt <i>optional</i>		Instant

JobDTO

Properties

Name	Description	Schema
id <i>optional</i>		integer (int64)
strategy <i>optional</i>		string
side <i>optional</i>		string
status <i>optional</i>		string
amount <i>optional</i>		number (double)
coins <i>optional</i>		< string > array
discountPct <i>optional</i>		number (double)
stepsDone <i>optional</i>		integer (int32)
stepsTotal <i>optional</i>		integer (int32)
durationMinut es <i>optional</i>		number (double)
createdAt <i>optional</i>		Instant

Name	Description	Schema
updatedAt <i>optional</i>		Instant

Void

User

Properties

Name	Description	Schema
id <i>optional</i>		string
email <i>optional</i>		string
username <i>optional</i>		string

Discord

Properties

Name	Description	Schema
discordId <i>optional</i>		string
username <i>optional</i>		string

Exchange

Properties

Name	Description	Schema
id <i>optional</i>		string
provider <i>optional</i>		string
name <i>optional</i>		string
readOnlyApiKey <i>optional</i>		string
readOnlyApiSecret <i>optional</i>		string
readWriteApiKey <i>optional</i>		string
readWriteApiSecret <i>optional</i>		string

UserConnectionsDTO

Properties

Name	Description	Schema
user <i>optional</i>		User
discord <i>optional</i>		Discord
exchangeClient <i>s optional</i>		< Exchange > array

UserRegisterRequestDTO

Properties

Name	Description	Schema
username <i>optional</i>		string
password <i>optional</i>		string
email <i>optional</i>		string
firstName <i>optional</i>		string
lastName <i>optional</i>		string
dateOfBirth <i>optional</i>		string

AuthRequestDTO

Properties

Name	Description	Schema
identifier <i>optional</i>		string
password <i>optional</i>		string

AuthResponseDTO

Properties

Name	Description	Schema
twoFactorRequired <i>optional</i>		boolean

Name	Description	Schema
user <i>optional</i>		UserDTO
token <i>optional</i>		string
expiration <i>optional</i>		string

OtpVerificationRequestDTO

Properties

Name	Description	Schema
email <i>optional</i>		string
code <i>optional</i>		string

UserUpdatePasswordRequestDTO

Properties

Name	Description	Schema
email <i>optional</i>		string

UserUpdatePasswordRequestVerifyDTO

Properties

Name	Description	Schema
code <i>optional</i>		string
email <i>optional</i>		string
newPassword <i>optional</i>		string

DiscordExchangeRequestDTO

Properties

Name	Description	Schema
code <i>optional</i>		string
codeVerifier <i>optional</i>		string

Source

Properties

Name	Description	Schema
title <i>optional</i>		string
region <i>optional</i>		string
domain <i>optional</i>		string
path <i>optional</i>		string
type <i>optional</i>		string

Currency

Votes

Properties

Name	Description	Schema
negative <i>optional</i>		integer (int32)
positive <i>optional</i>		integer (int32)
important <i>optional</i>		integer (int32)
liked <i>optional</i>		integer (int32)
disliked <i>optional</i>		integer (int32)
lol <i>optional</i>		integer (int32)
toxic <i>optional</i>		integer (int32)
saved <i>optional</i>		integer (int32)
comments <i>optional</i>		integer (int32)

CryptoPanicPost

Properties

Name	Description	Schema
kind <i>optional</i>		string
domain <i>optional</i>		string
source <i>optional</i>		Source
title <i>optional</i>		string

Name	Description	Schema
published_at <i>optional</i>		string
slug <i>optional</i>		string
currencies <i>optional</i>		< Currency > array
id <i>optional</i>		integer (int64)
url <i>optional</i>		string
created_at <i>optional</i>		string
votes <i>optional</i>		Votes

CryptoPanicFrontendResponse

Properties

Name	Description	Schema
count <i>optional</i>		integer (int32)
results <i>optional</i>		< CryptoPanicPost > array

PushNotification

Properties

Name	Description	Schema
id <i>optional</i>		integer (int64)
userId <i>optional</i>		integer (int64)
read <i>optional</i>		boolean
category <i>optional</i>		enum (SYSTEM,MARKETING,USER,JOB,MARKET,HEALTH)
type <i>optional</i>		enum (INFO,WARNING)
title <i>optional</i>		string
body <i>optional</i>		string
data <i>optional</i>		string

Name	Description	Schema
channelId <i>optional</i>		string
notificationLevel <i>optional</i>		integer (int32)
deepLink <i>optional</i>		string
sentAt <i>optional</i>		Instant

CurrencyPair

Properties

Name	Description	Schema
base <i>optional</i>		Currency
counter <i>optional</i>		Currency

InstrumentInfoDTO

Properties

Name	Description	Schema
priceScale <i>optional</i>		integer (int32)
quantityStep <i>optional</i>		BigDecimal
timestamp <i>optional</i>		integer (int64)

MarketDataDTO

Properties

Name	Description	Schema
instrument <i>optional</i>		CurrencyPair
currentPrice <i>optional</i>		BigDecimal
change24h <i>optional</i>		BigDecimal
high24h <i>optional</i>		BigDecimal

Name	Description	Schema
low24h <i>optional</i>		BigDecimal
price24hAgo <i>optional</i>		BigDecimal
price1hAgo <i>optional</i>		BigDecimal
volume24h <i>optional</i>		BigDecimal
openInterestVa lue <i>optional</i>		BigDecimal
fundingRate <i>optional</i>		BigDecimal
nextFundingTi me <i>optional</i>		integer (int64)
instrumentInfo <i>optional</i>		InstrumentInf oDTO